Resilient Communication Networks

Designing Resilient Communication Networks

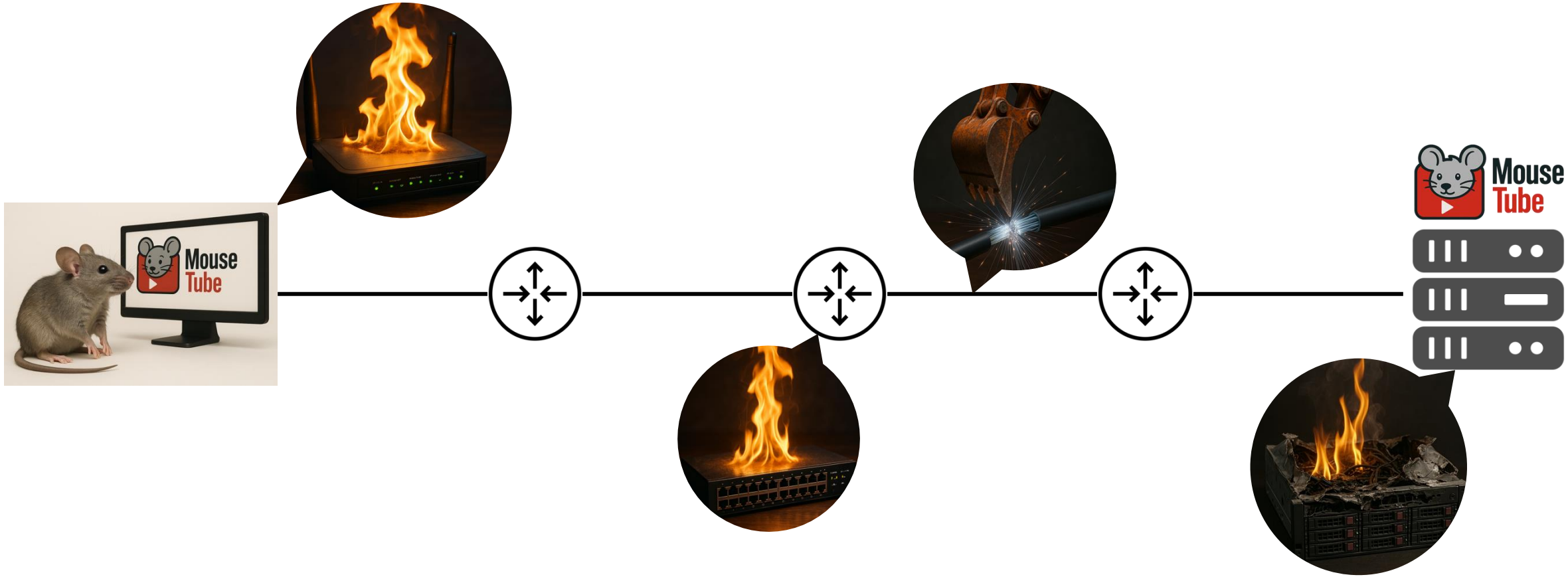
Redundancy Mechanisms



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Dept. of Computer Science (adjunct Professor)
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Disclaimer: some of these slides have been derived from Günter Schäfer's lecture „Protection of Communication Infrastructures", Mathias Fischers Lecture „Resilient Networks", Matthias Hollick's Lecture „ Ringvorlesung KRITIS", and Ralf Steinmetz's Lecture "Communication Networks II"

Let‘s say we operate a streaming service MouseTube

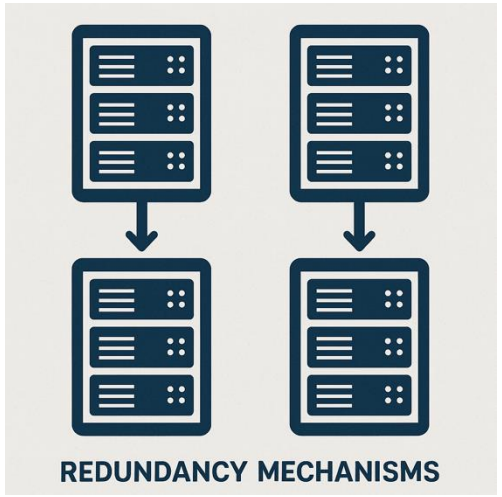


Enablers of Resilience in Communication Networks

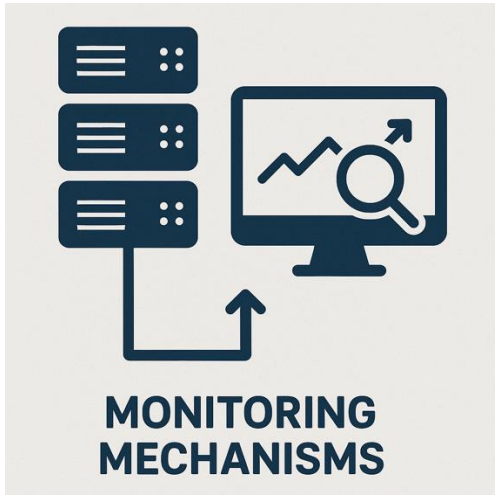
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Four general classes of enablers

- Redundancy for robustness against disruptions
- Protection for robustness against attacks
- Monitoring for detecting disruptions
- Recovery from failures

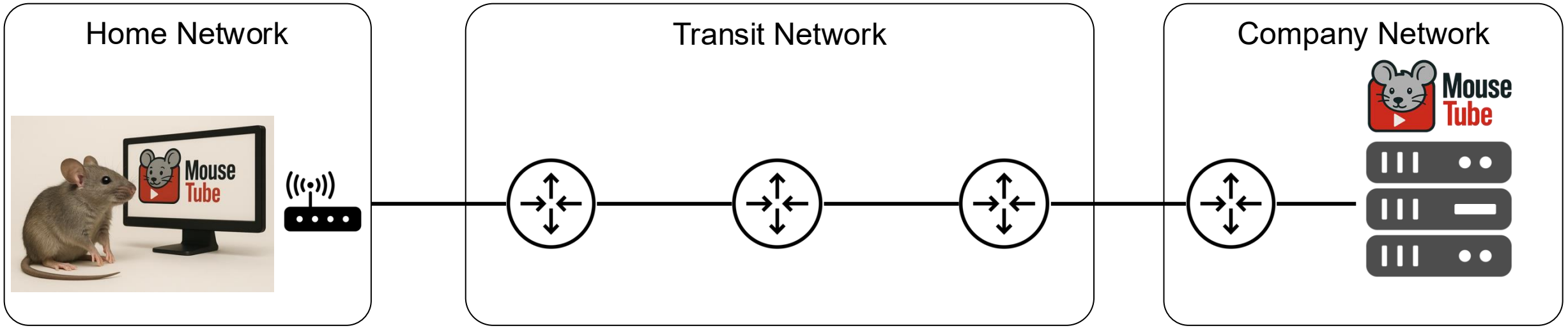








What components would need to be redundant?



Different Types of Redundancy Mechanisms

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Redundancy of paths

- Router/Switch redundancy
- Link redundancy

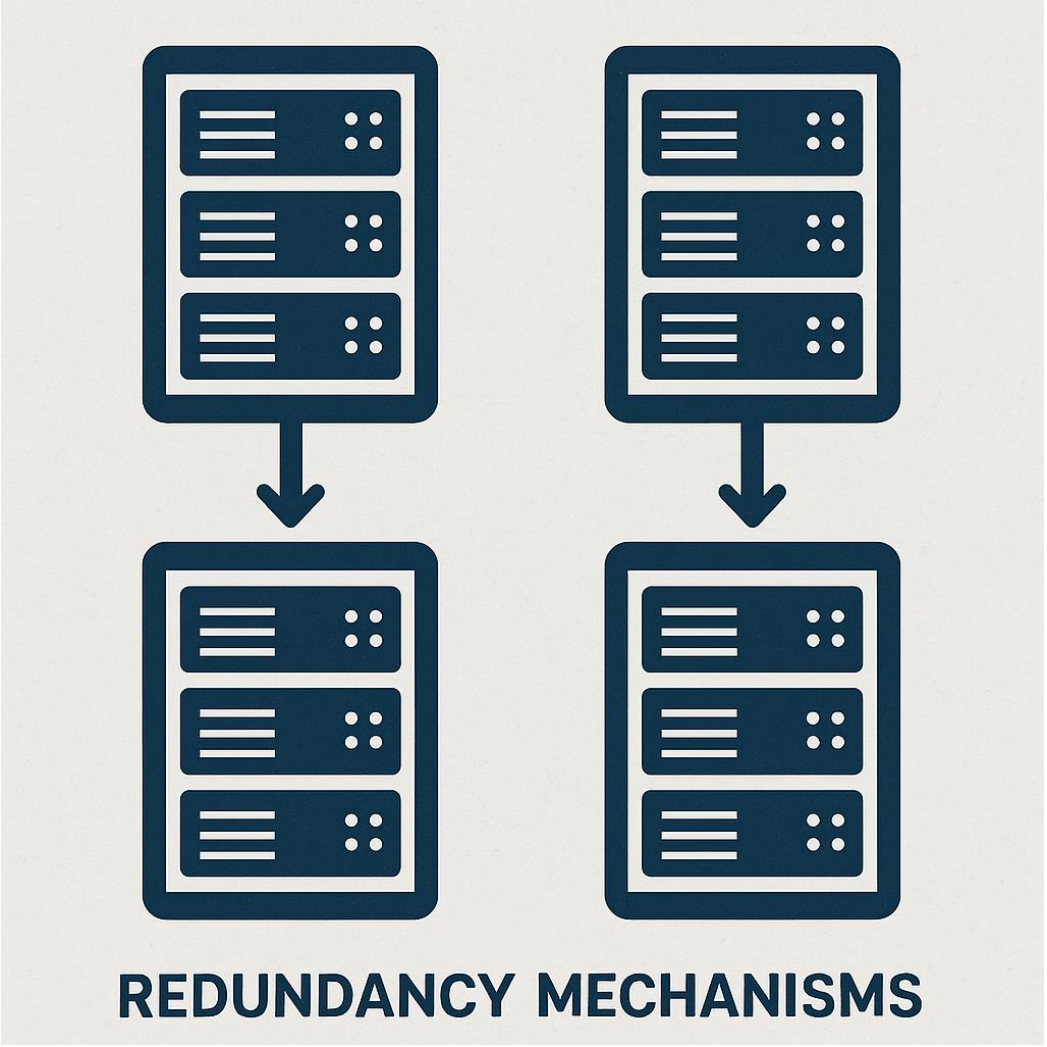
Redundancy of computation

- Multi-server operation

Redundancy of storage

- Database synchronization

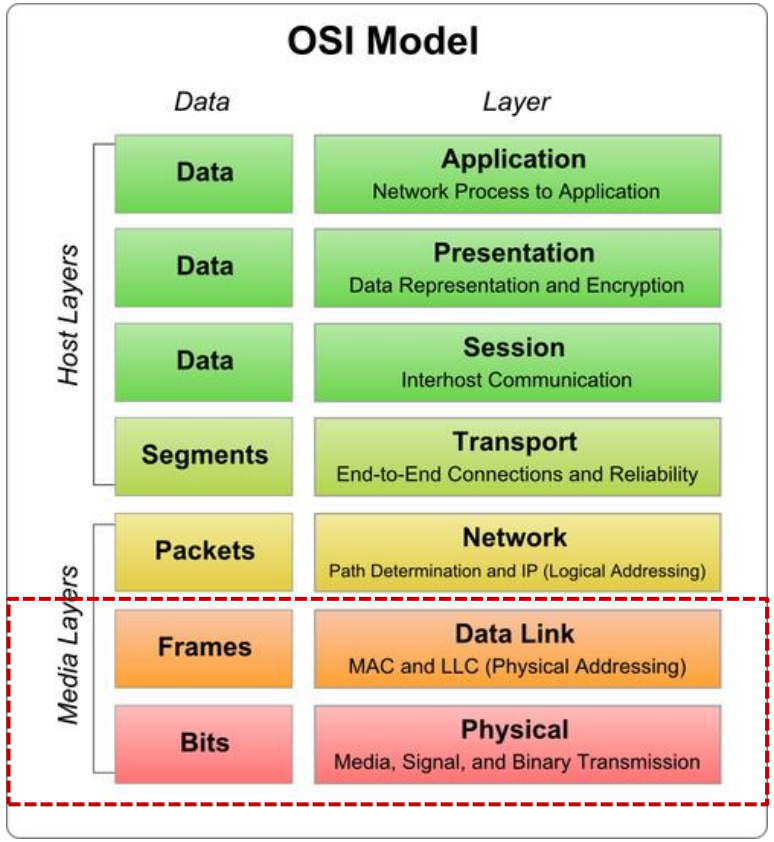
How are these types of redundancy implemented?



Path Redundancy

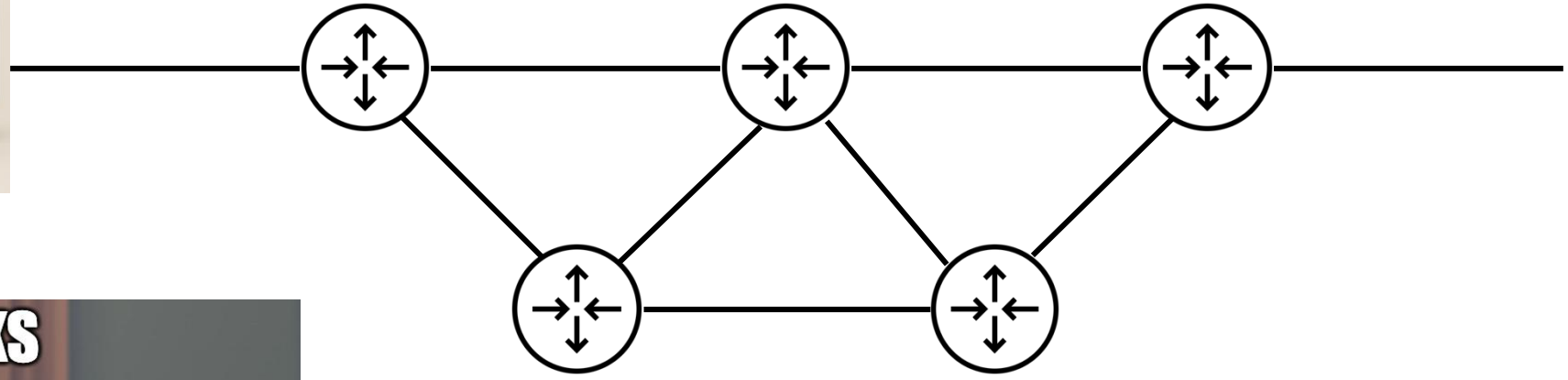
How to utilize redundant paths?

Link Redundancy – Ethernet



Link Redundancy – Ethernet









Loops in your network will cause issues without mechanisms like STP

In the simplest approach, loops are just detected and removed

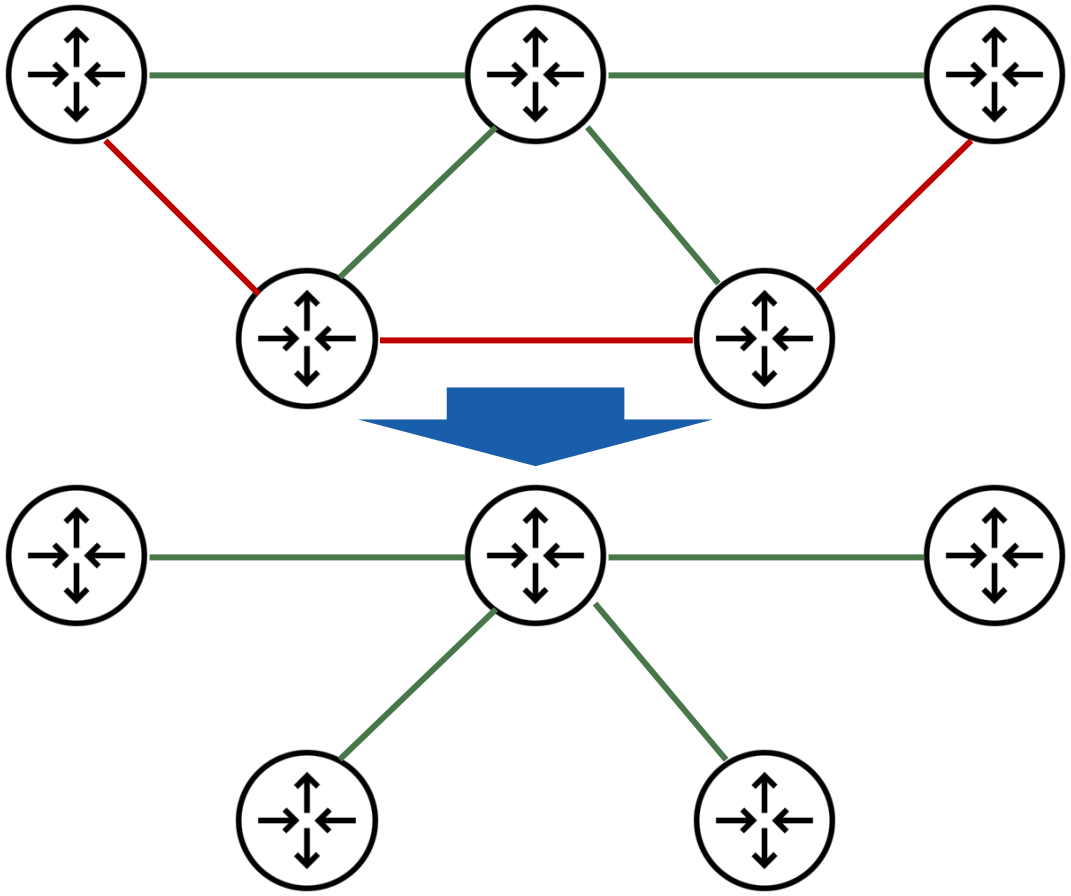
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Especially in small networks, the Spanning Tree Protocol (STP) is used

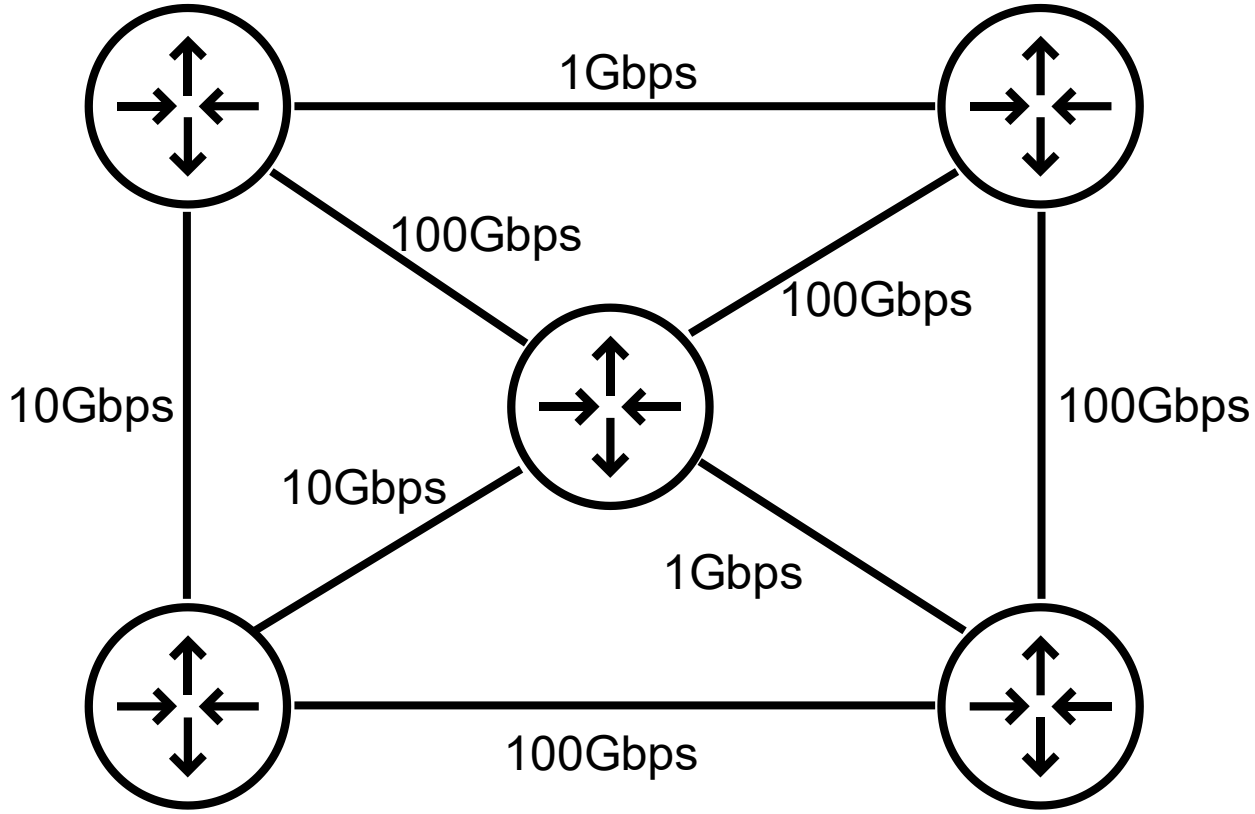
- Finds a loop-free logical topology of an Ethernet network
- Additional links are deactivated to prevent loops
- Only one of the links is used to avoid loops

Includes recovery after link failure

- Failure recovery is generally slow (tens of seconds)
- Rapid Spanning Tree Protocol (RSTP) allows for fast recovery (few seconds)
  - Redundant links cannot be used simultaneously
  - No possibility for multipath
  - Might still be too slow for some applications



Spanning Tree Protocol (STP)
Example

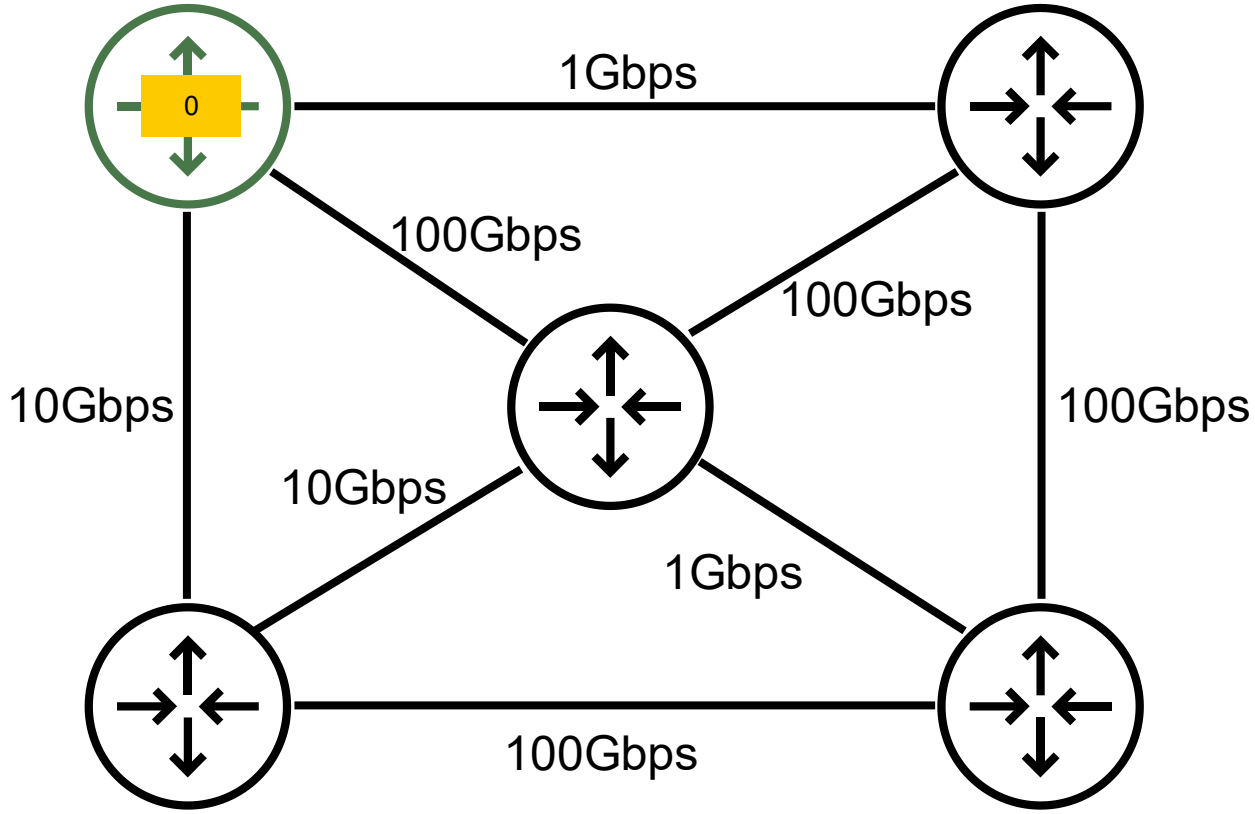


Spanning Tree Protocol (STP)
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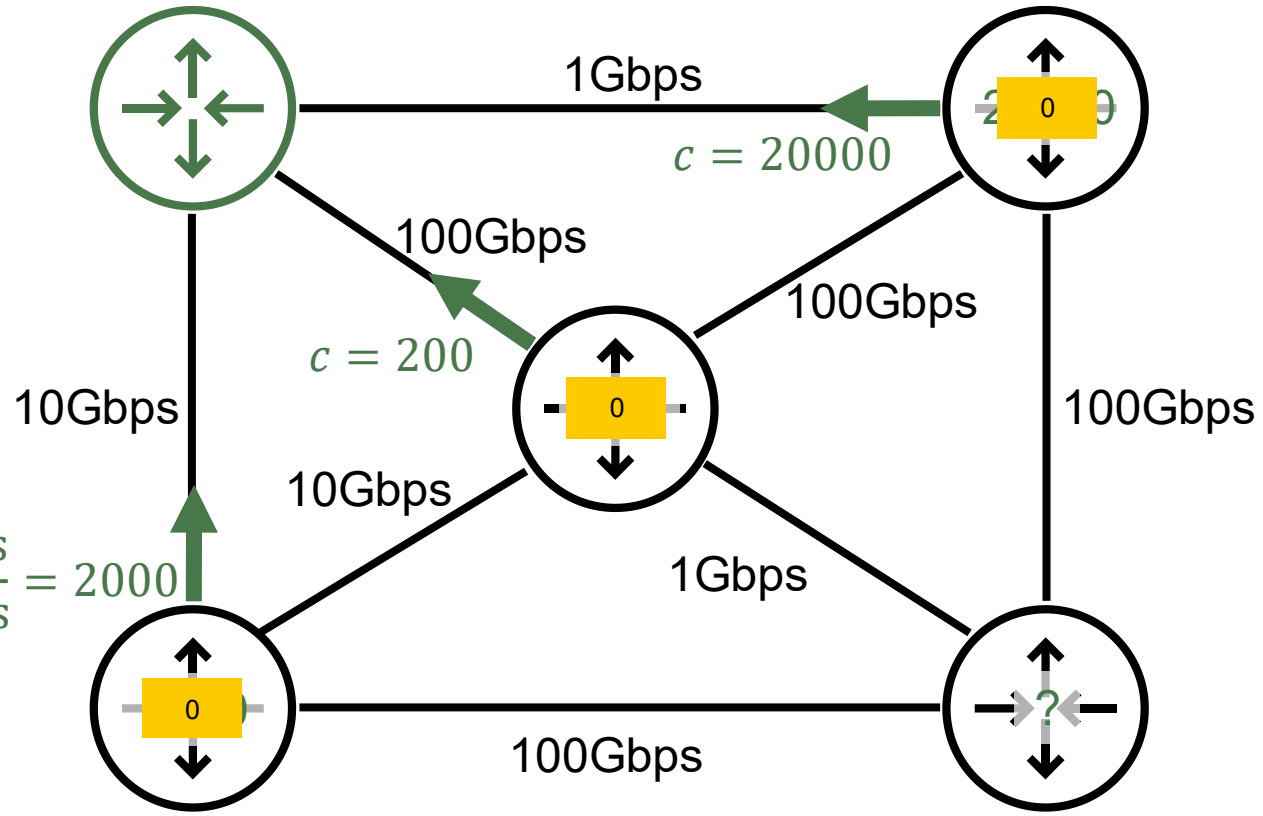
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   - Often based on data rate: $\frac{20\text{Tbps}}{\text{bandwidth}}$
   - Calculated based on the path costs of the neighbors

Data rate	Cost
1Gbps	20000
10Gbps	2000
100Gbps	200

$$c = \frac{20\text{Tbps}}{10\text{Gbps}} = 2000$$

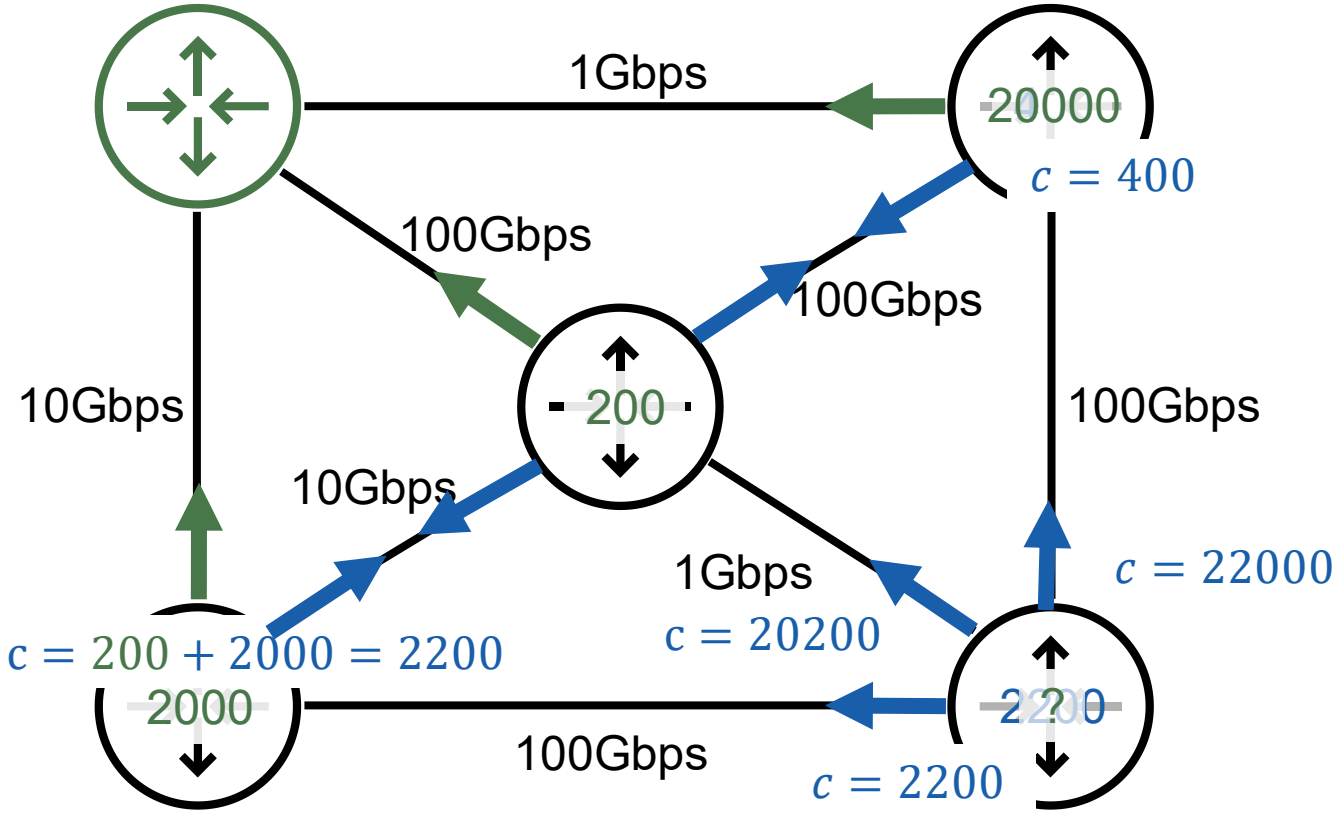


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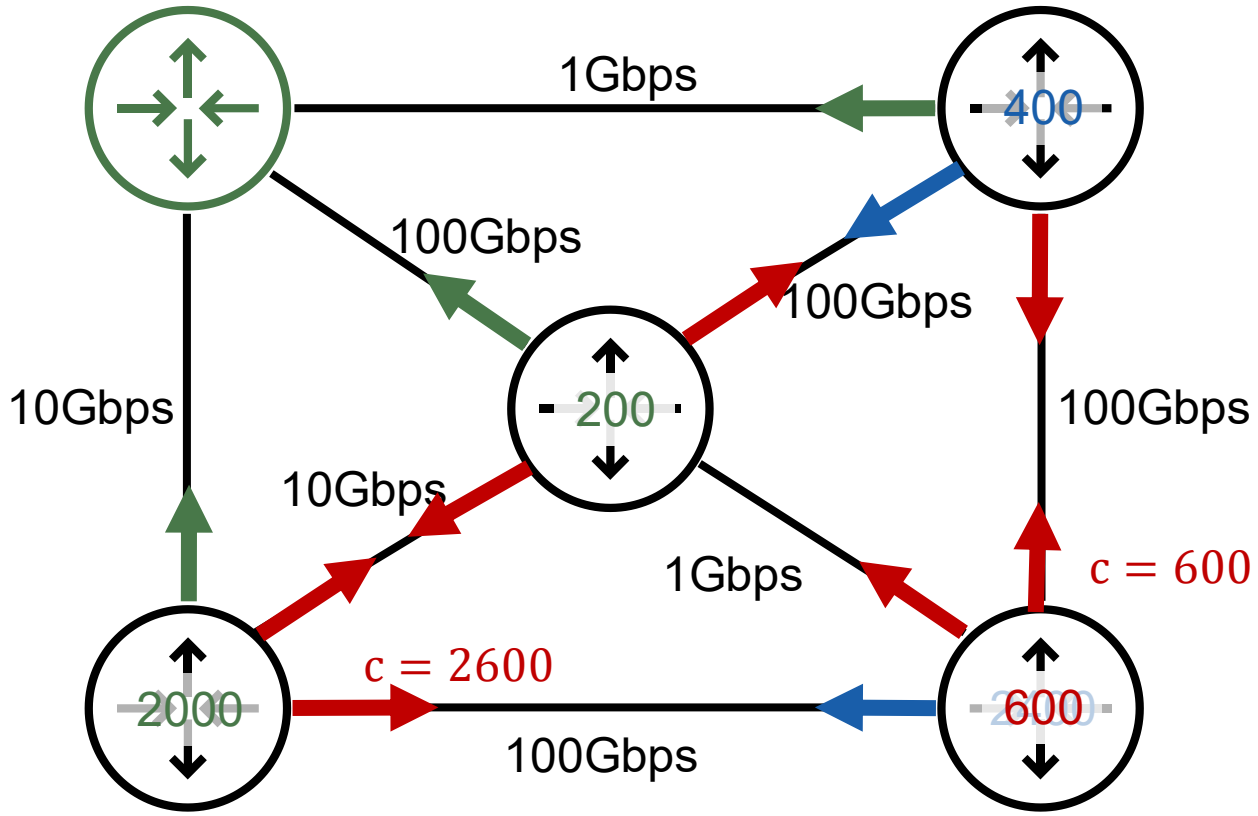


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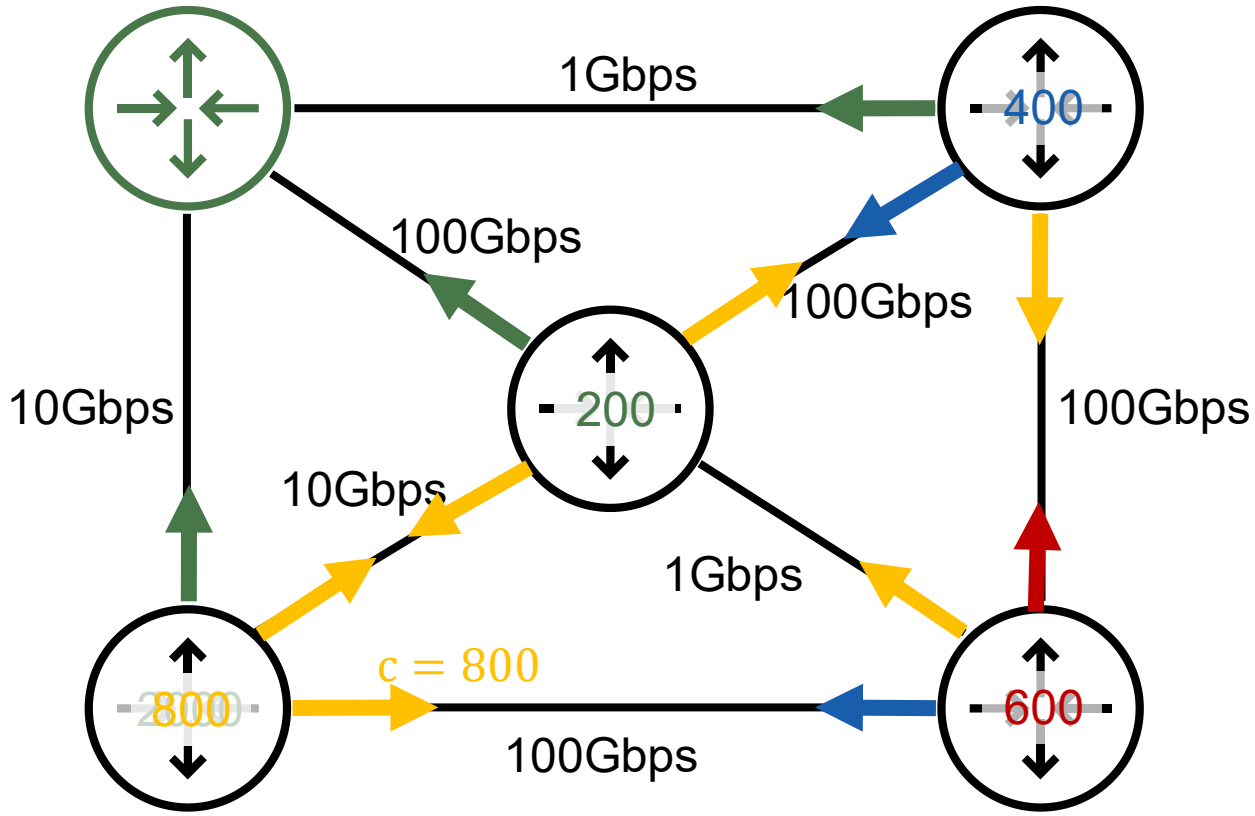


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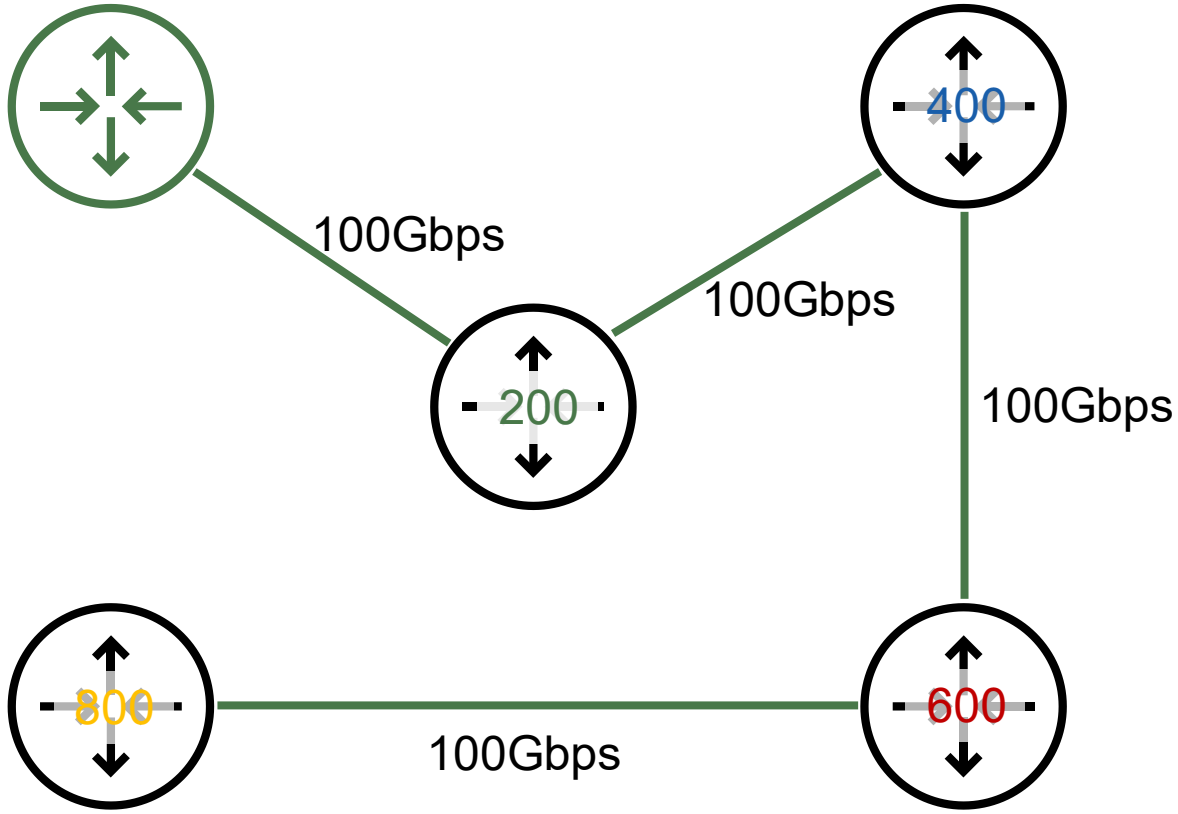
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   - To avoid loops, each port on a switch goes through several states, e.g.,
     - Blocking: Port is blocked by STP
     - Listening: Waiting for additional info
     - Learning: MAC-Learning only
     - Forwarding: Normal operation



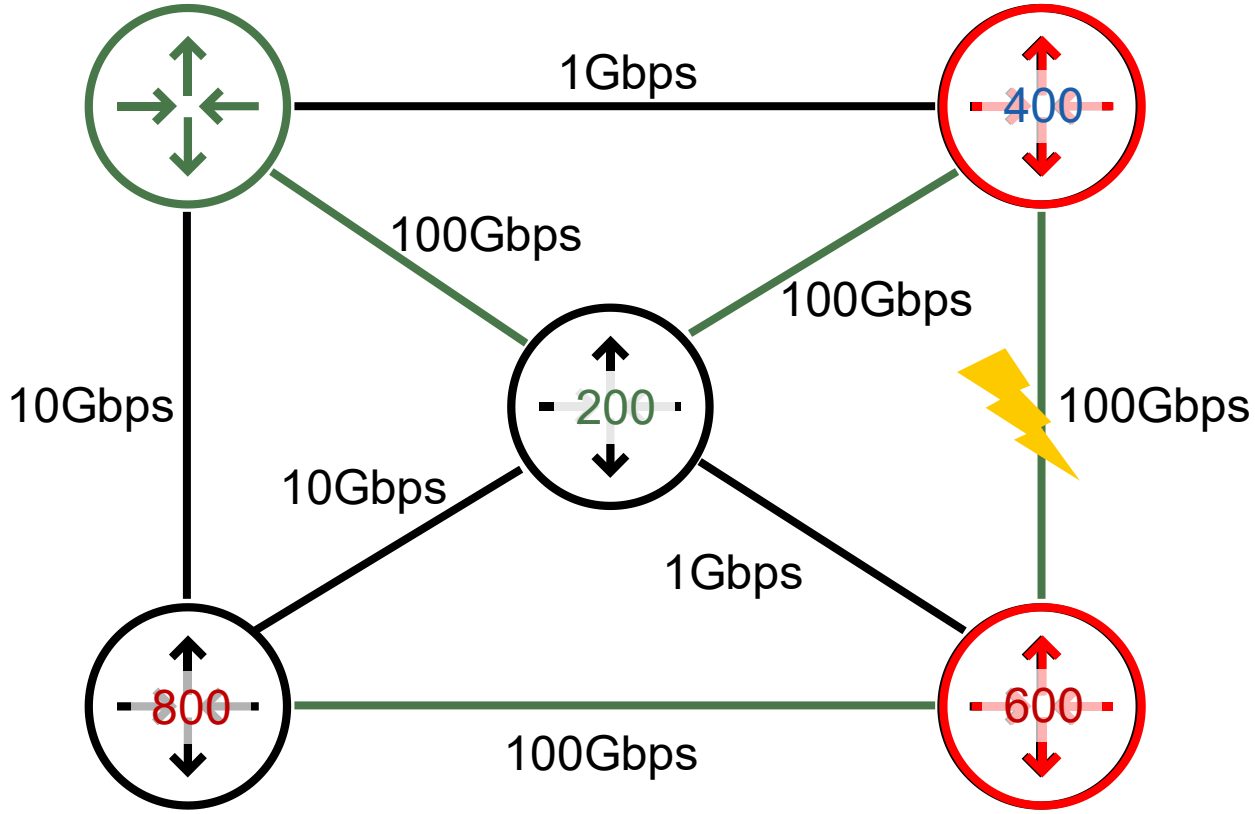
Spanning Tree Protocol (STP)
Example of Link Failure

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If a link fails, STP will eventually establish a new tree

- Directly affected bridges switch to listening (do not forward packets anymore)
- The affected bridges send a topology change notification (TCN) to the root bridge

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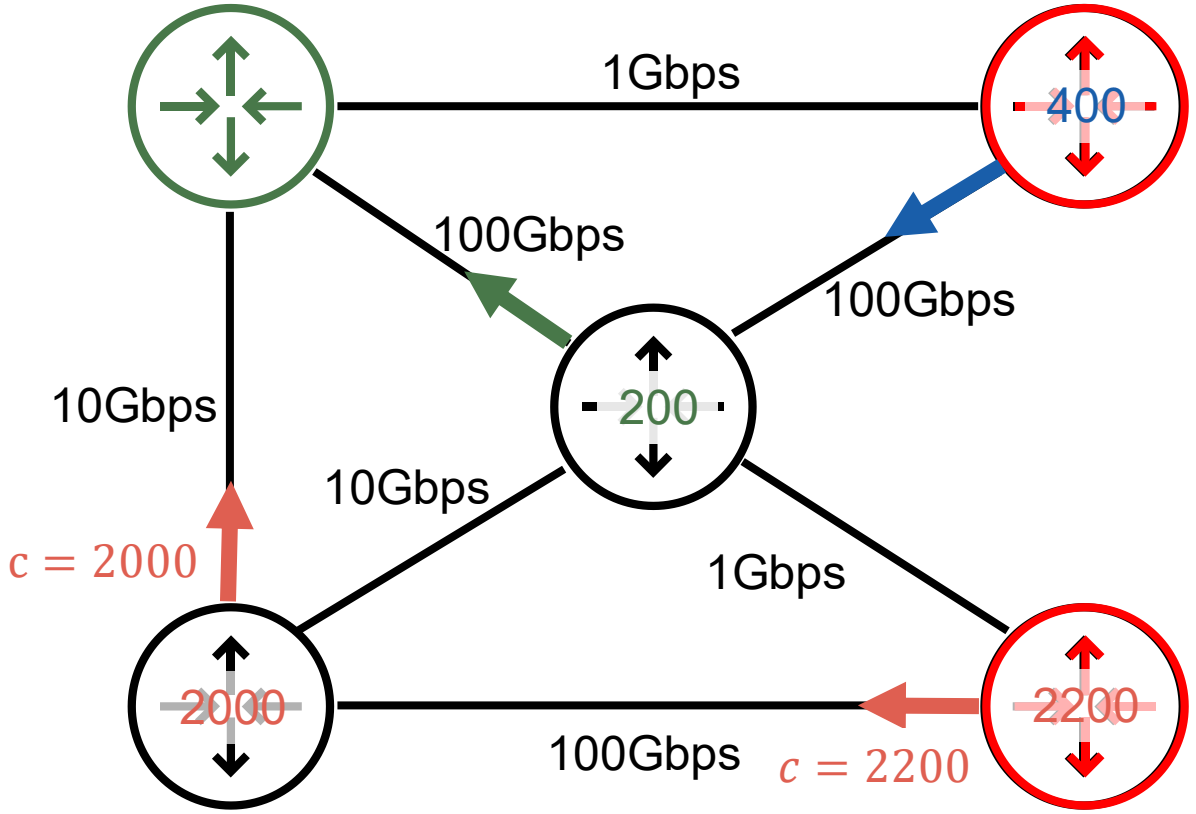
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Non-connected bridges will timeout and choose a different path to the root

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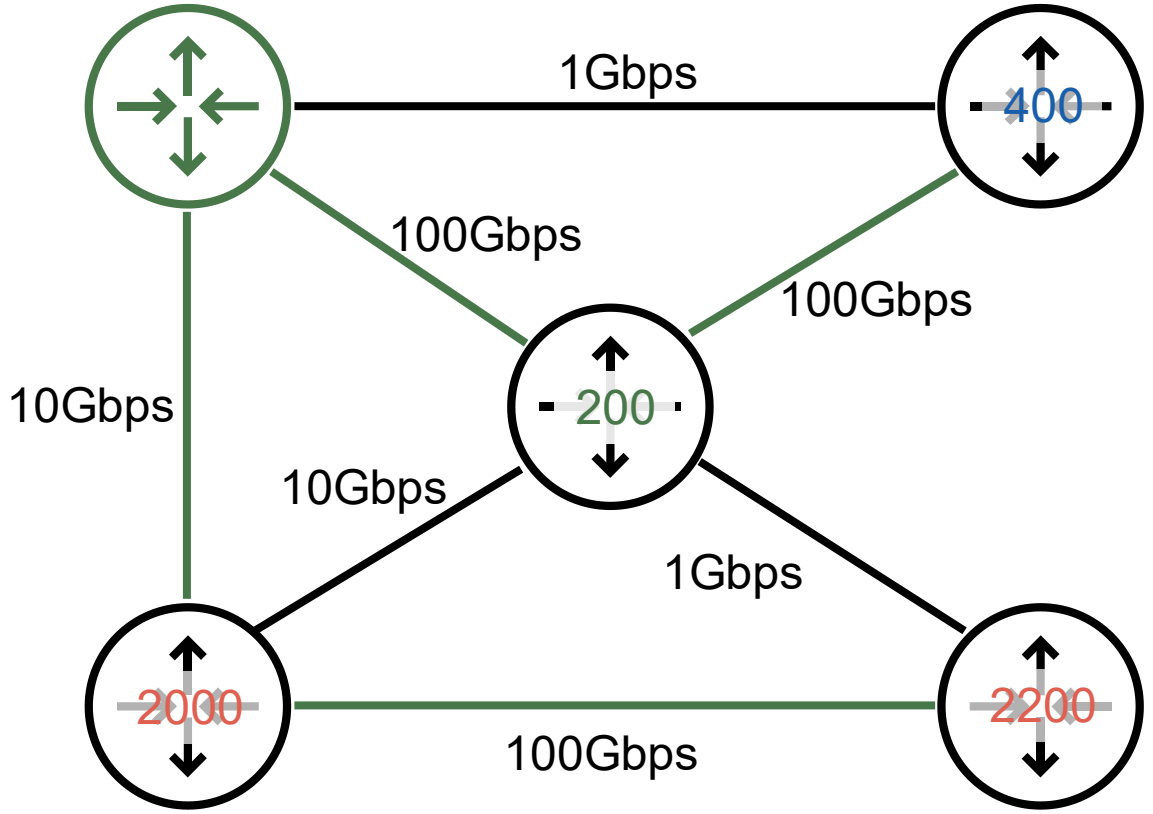
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Rapid Spanning Tree Protocol (RSTP)

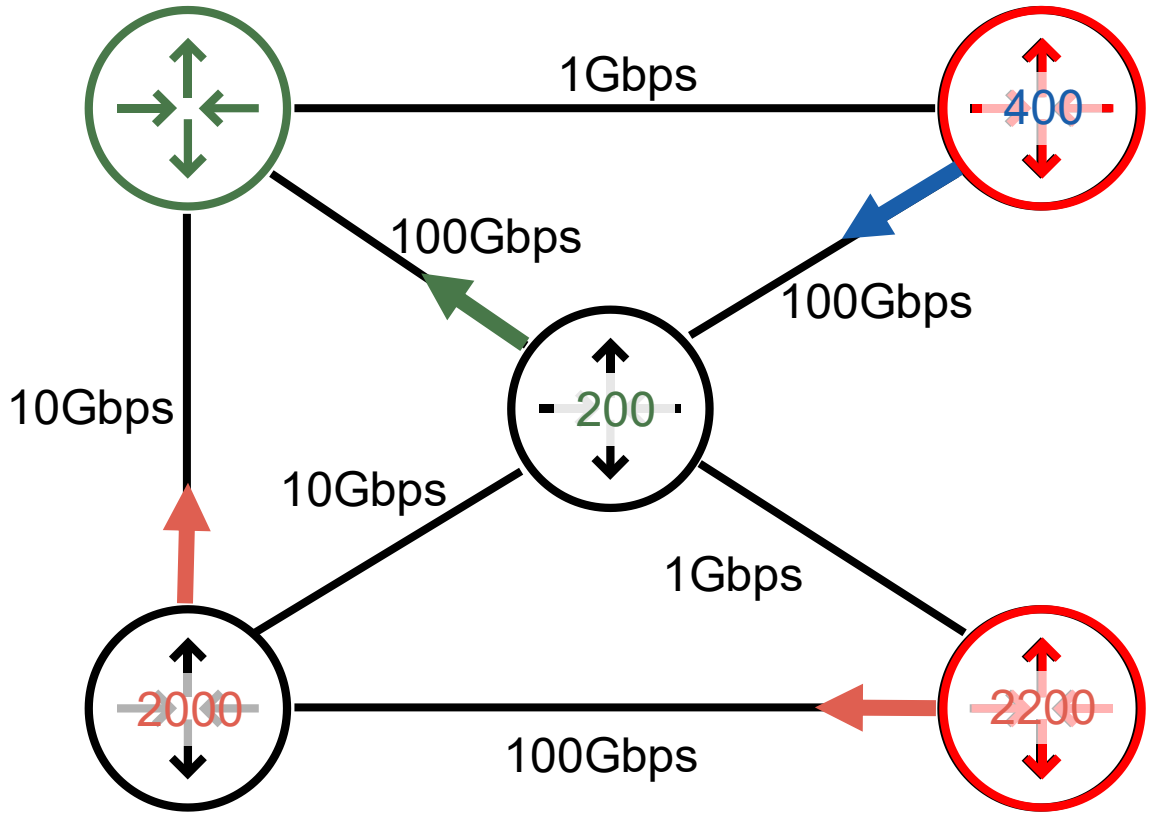
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Similar concept to STP with backwards compatibility

- Improves failure detection through regular heartbeats (Bridge Protocol Data Unit are sent by all bridges, not just by the root bridge)
- Allows for faster discarding of paths
- Improves failure handling through the calculation of backup paths

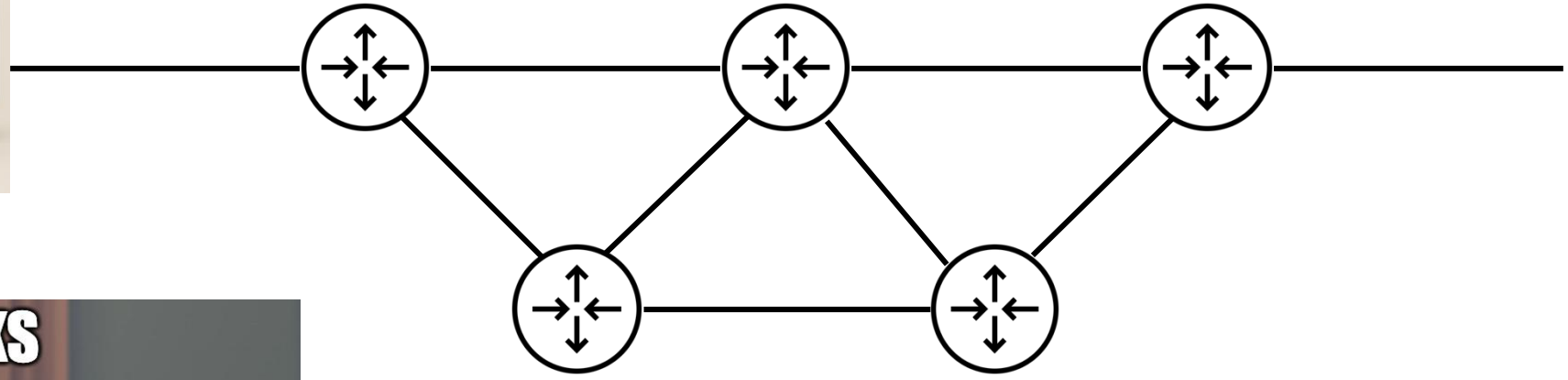
Separate handling of so-called edge ports

- Ports to which edge devices like printers and computers are connected
- Edge ports are always enabled and forwarding, as no routing loops can be created there



Link Redundancy – Ethernet









Shortest Path Bridging

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What if you actually want to use all available paths?

Shortest Path Bridging (SPB) has been developed to replace Spanning Tree Protocol (STP)

- SPB allows for the utilization of multiple paths between source and destination, while ensuring a loop-free layer 2 topology
- SPB improves link utilization and resilience compared to STP

SPB utilizes a routing protocol to find shortest paths between nodes in a network

- In the standard, the Link-State Interior Gateway Protocol (IS-IS) is utilized
- Differentiation between core switches (only connected to other switches) and edge switches

Shortest Path Bridging is somewhat similar to **Spanning Tree Protocol**, but each switch in SPB is the source of a tree

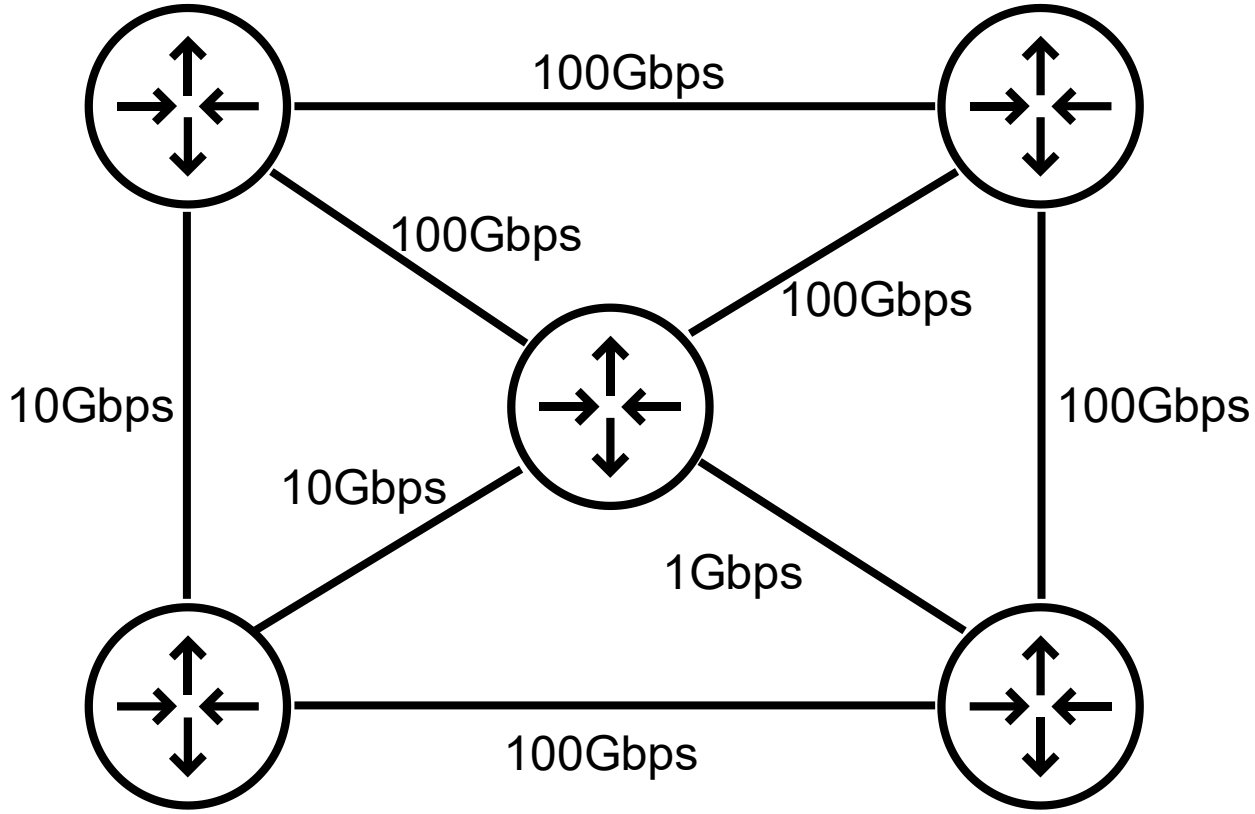
Back to the topology used for STP (with different link speeds)

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Find shortest paths from all switches to all other switches

- E.g., based on the cost metric from STP:

$$c = \frac{20\text{Tbps}}{\text{bandwidth}}$$



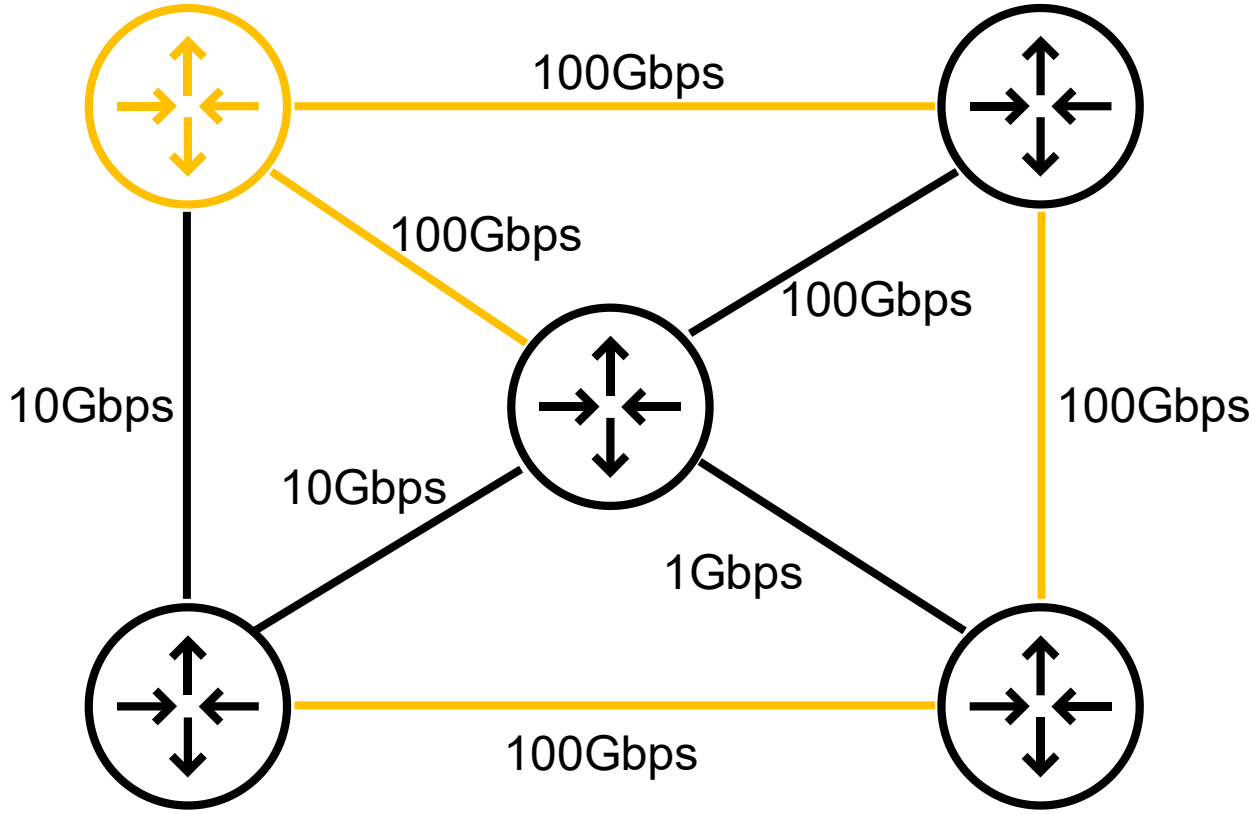
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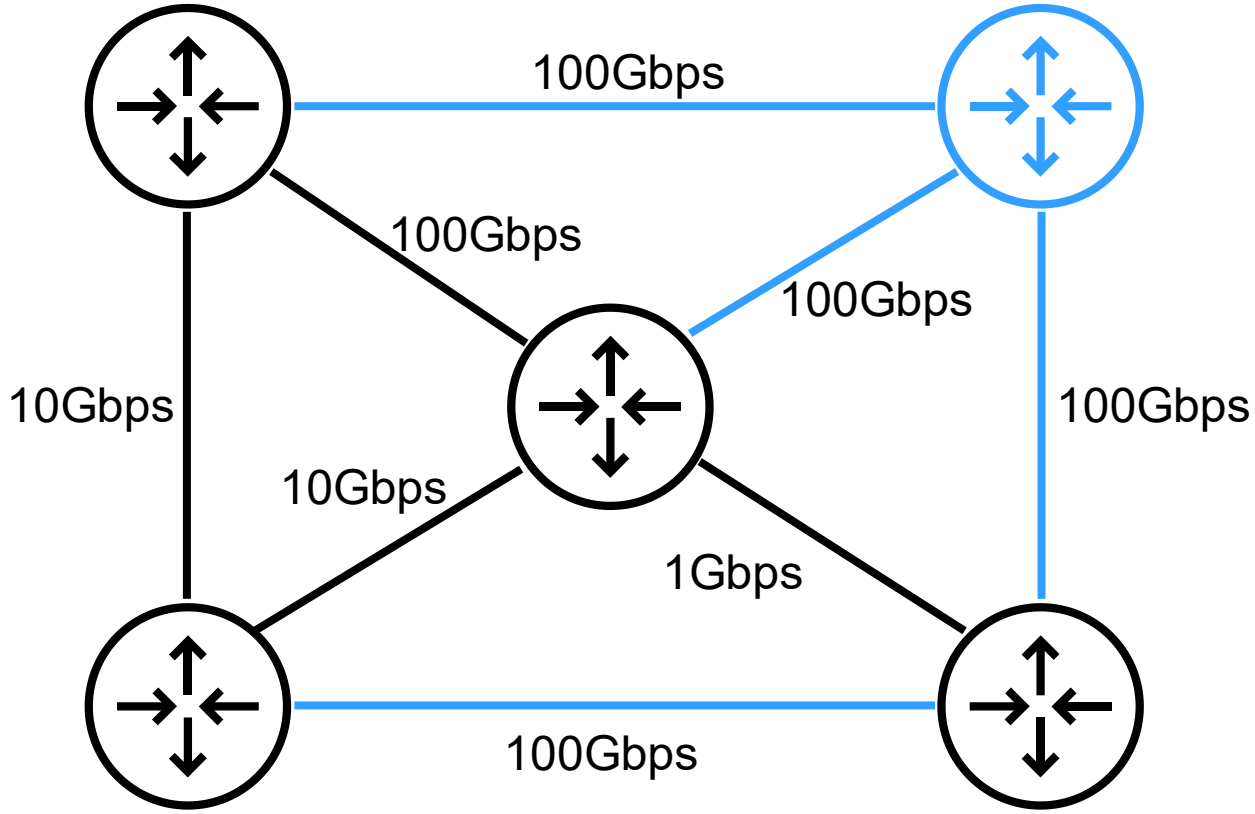
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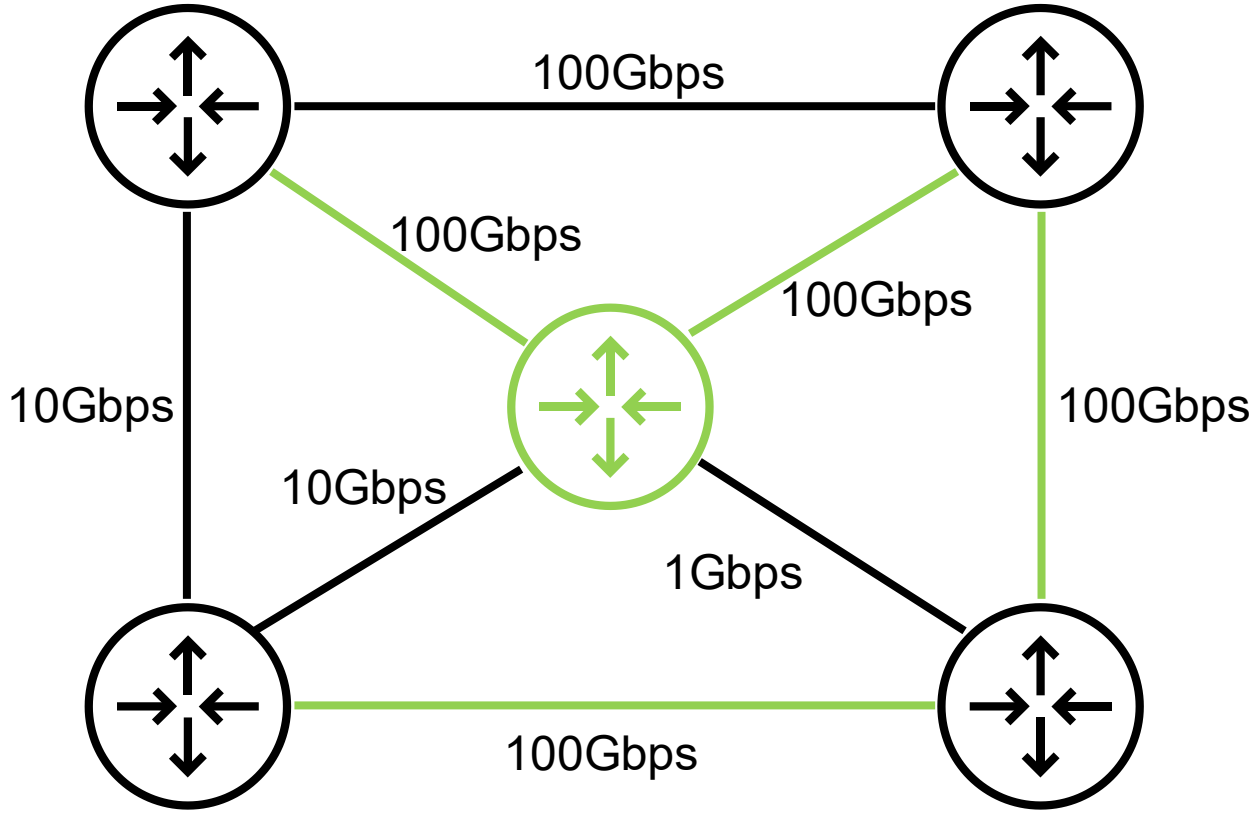
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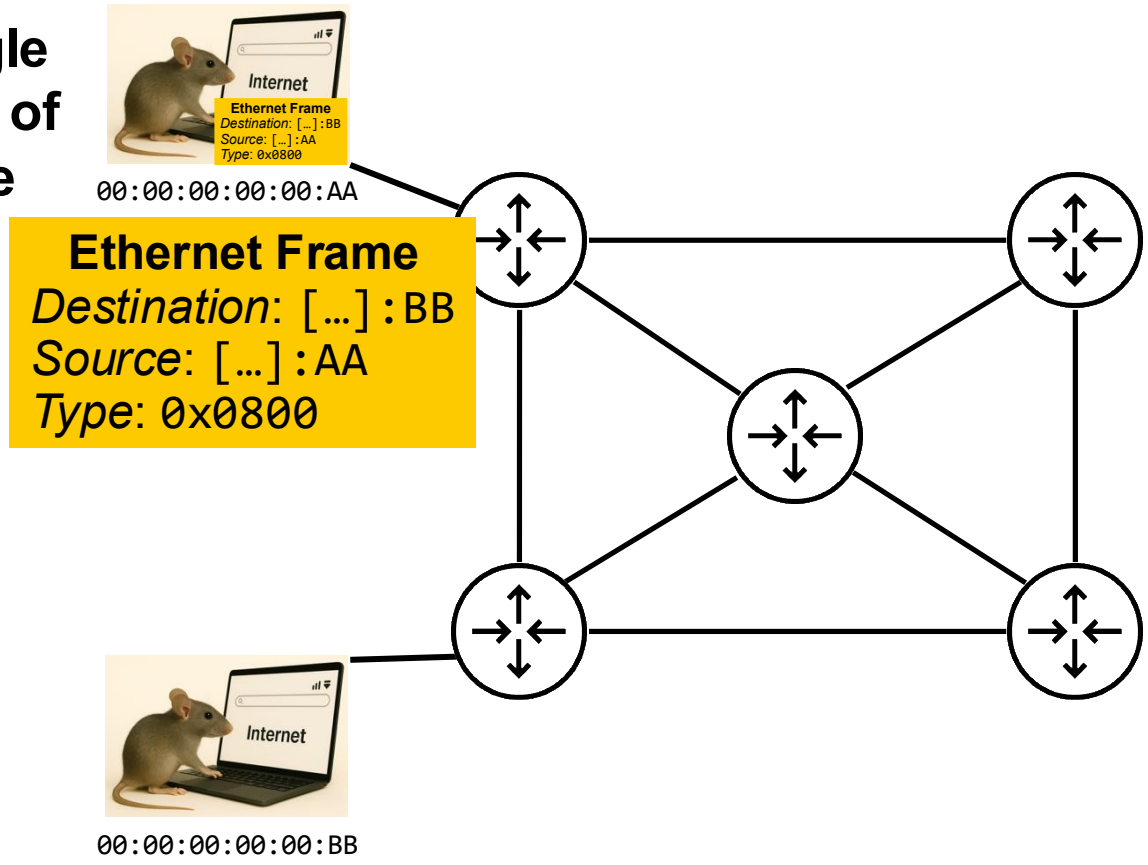


Shortest Path Bridging

Shortest Path Bridging is somewhat similar to **Spanning Tree Protocol**, but each switch in SPB is the source of a tree

Instead of utilizing the forwarding tree of a single network-wide root switch, the forwarding tree of the edge switch at which the packet enters the backbone network (root switch) is used

- Packets are forwarded away from the root switch
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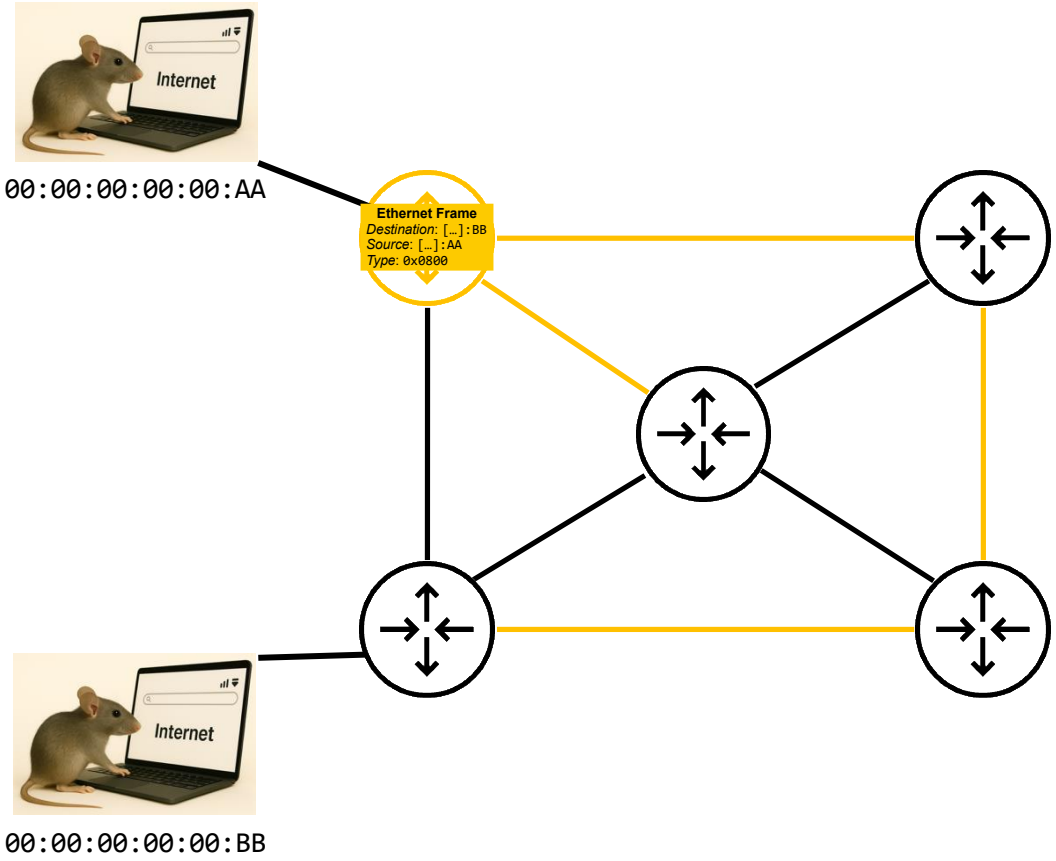


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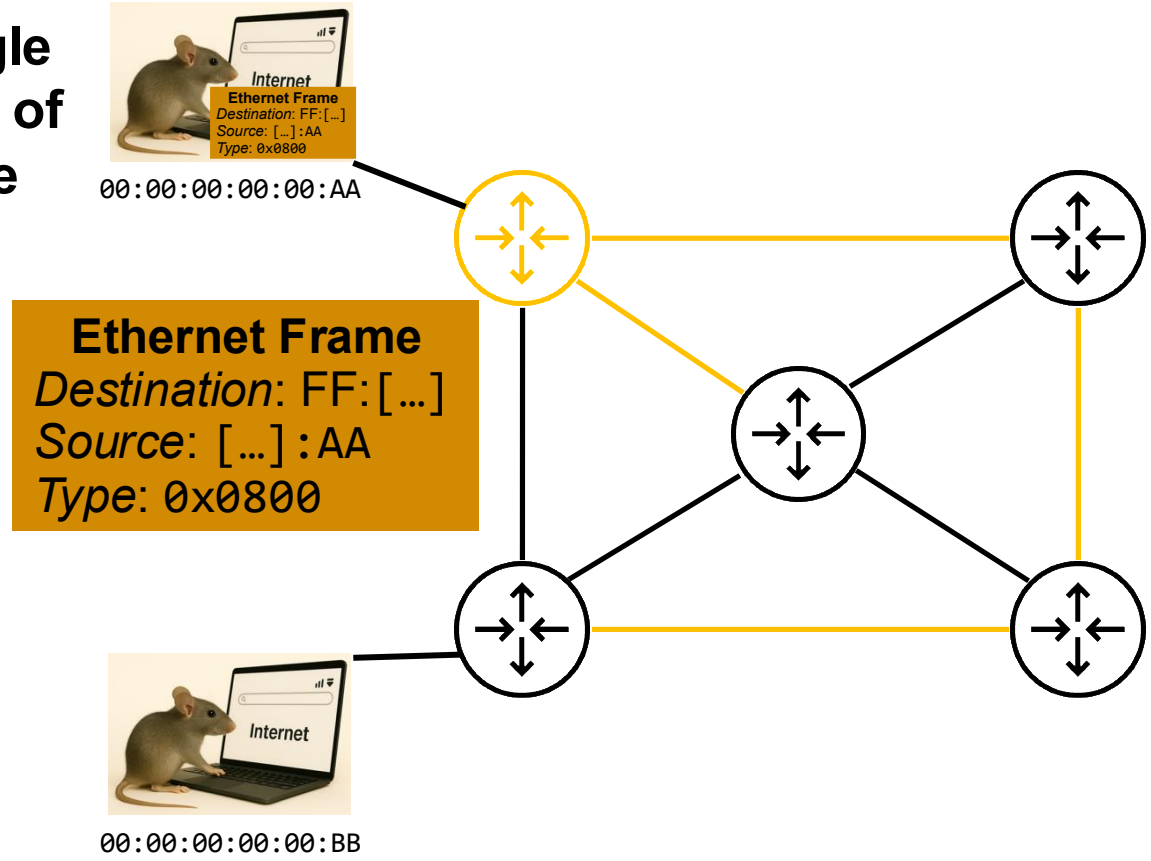


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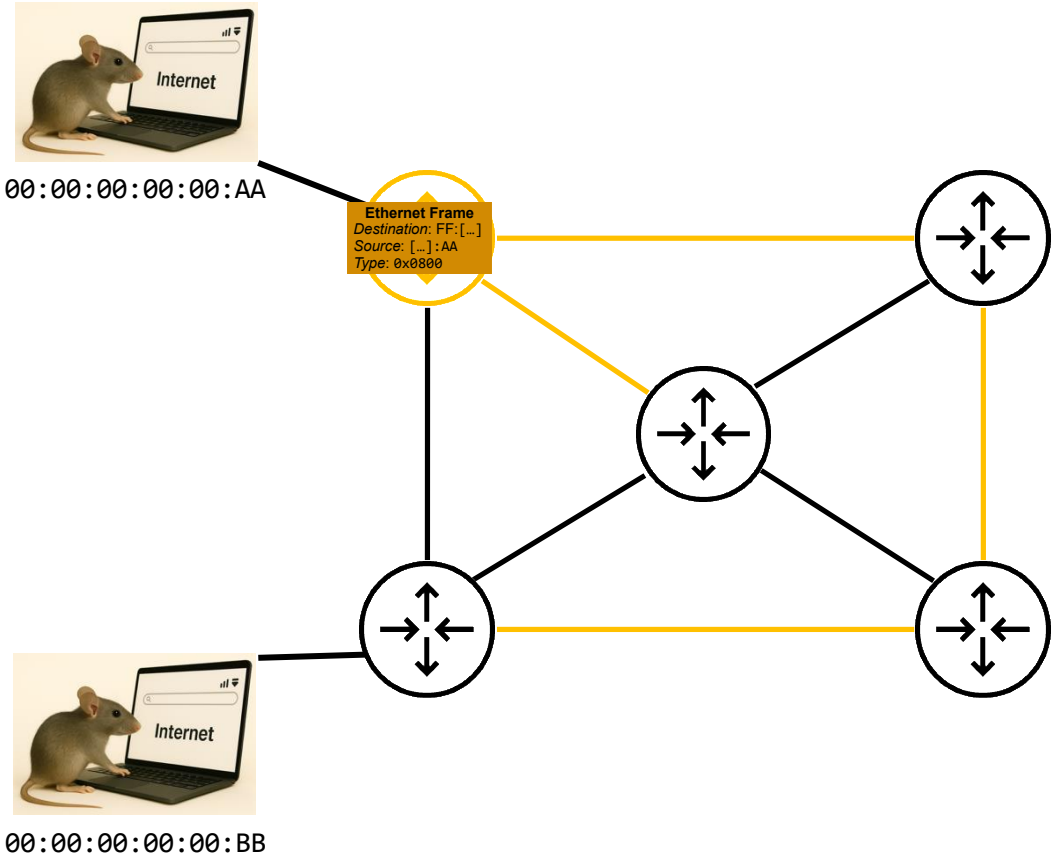


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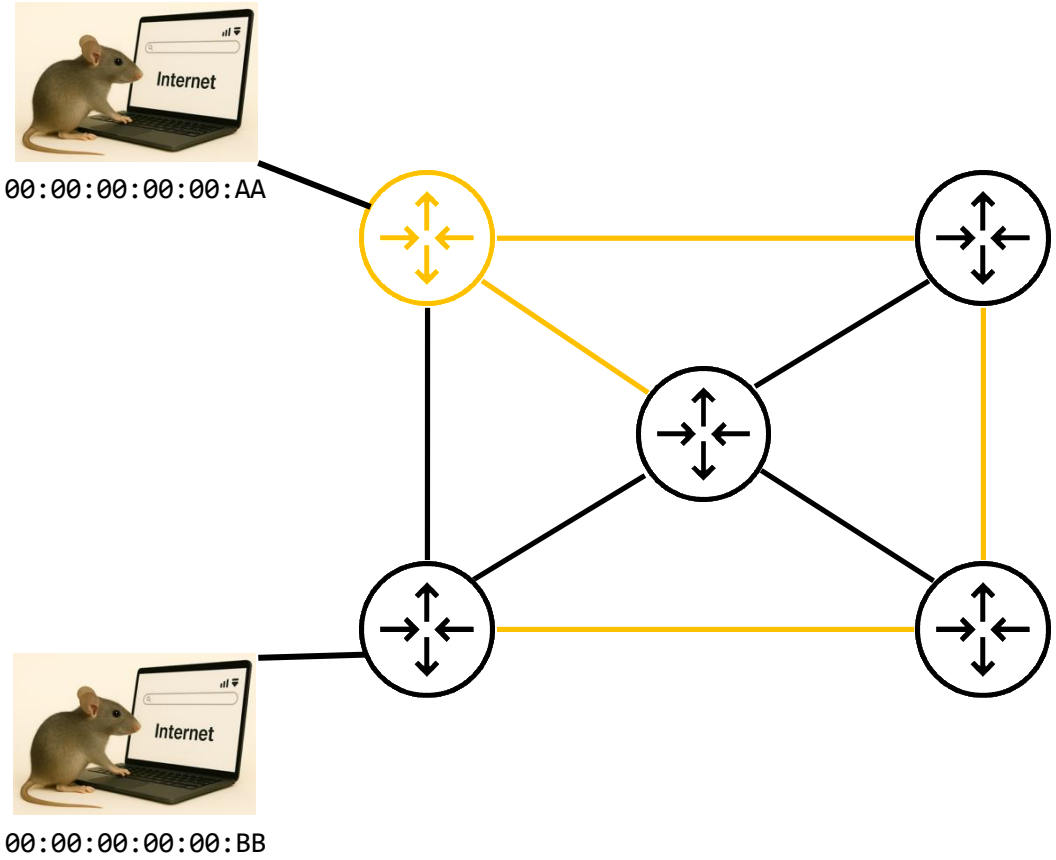
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How do you know the root switch of an Ethernet frame?

Two main options

- Shortest Path Bridging-VID (SPBV)
- Shortest Path Bridging-MAC (SPBM)



Shortest Path Bridging-MAC (SPBM)

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More common than Shortest Path Bridging-VID (SPBV)

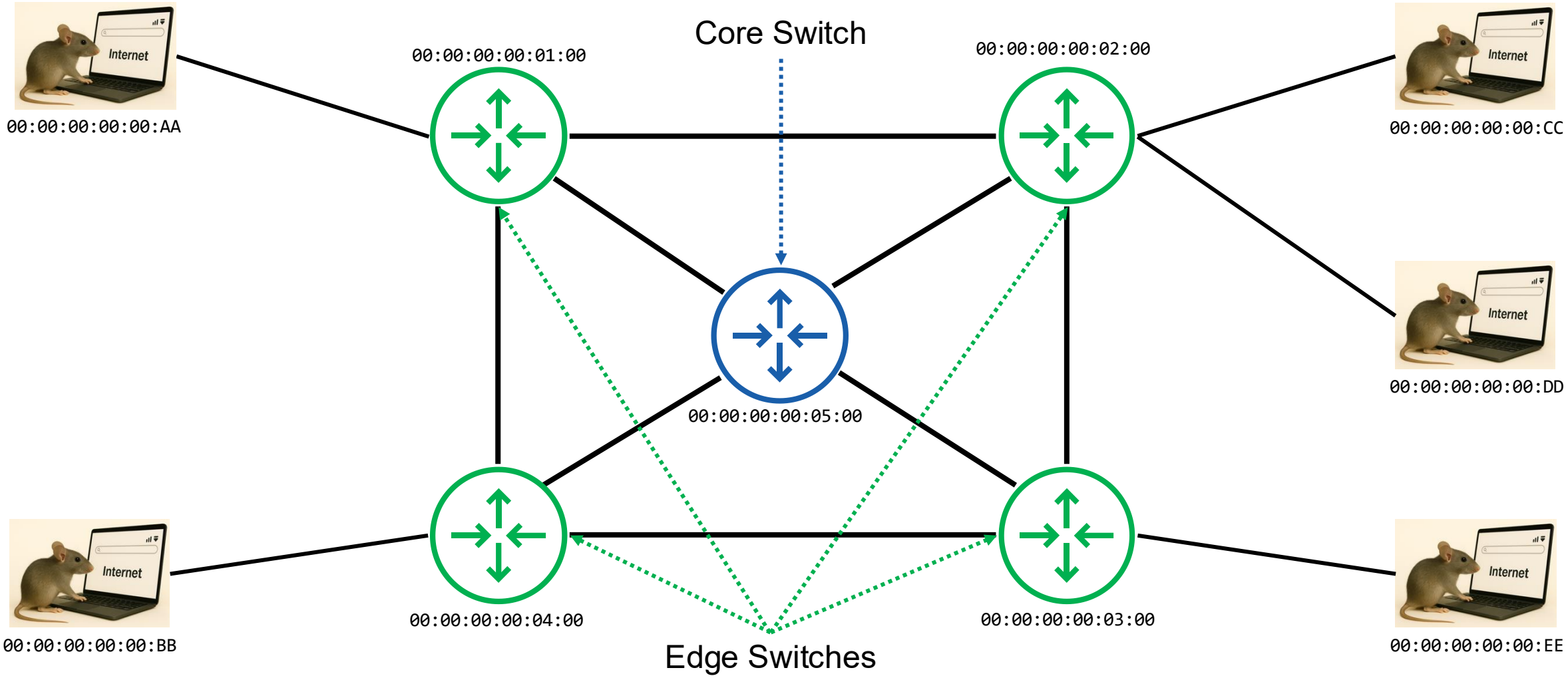
- SPBM is more scalable due to a higher number of identifiers (4096 for SPBV, aprox. 16M for SPBM)
- SPBV is backwards compatible with (M)STP, making it usable in legacy networks
- Frames are forwarded based on
  - the MAC-address of the edge switch (the first switch that the frame is received at)
  - the service ID *(we will come to that later)*
- For that, the frames are encapsulated in Ethernet frames themselves

Differentiation between Backbone Switches and Edge Switches

- Backbone switches are only connected to other switches
- Edge switches are connected to clients and other switches

Only the edge switches need to learn client MAC addresses

Shortest Path Bridging-MAC (SPBM)



Services in SPB

Networks often accommodate different services

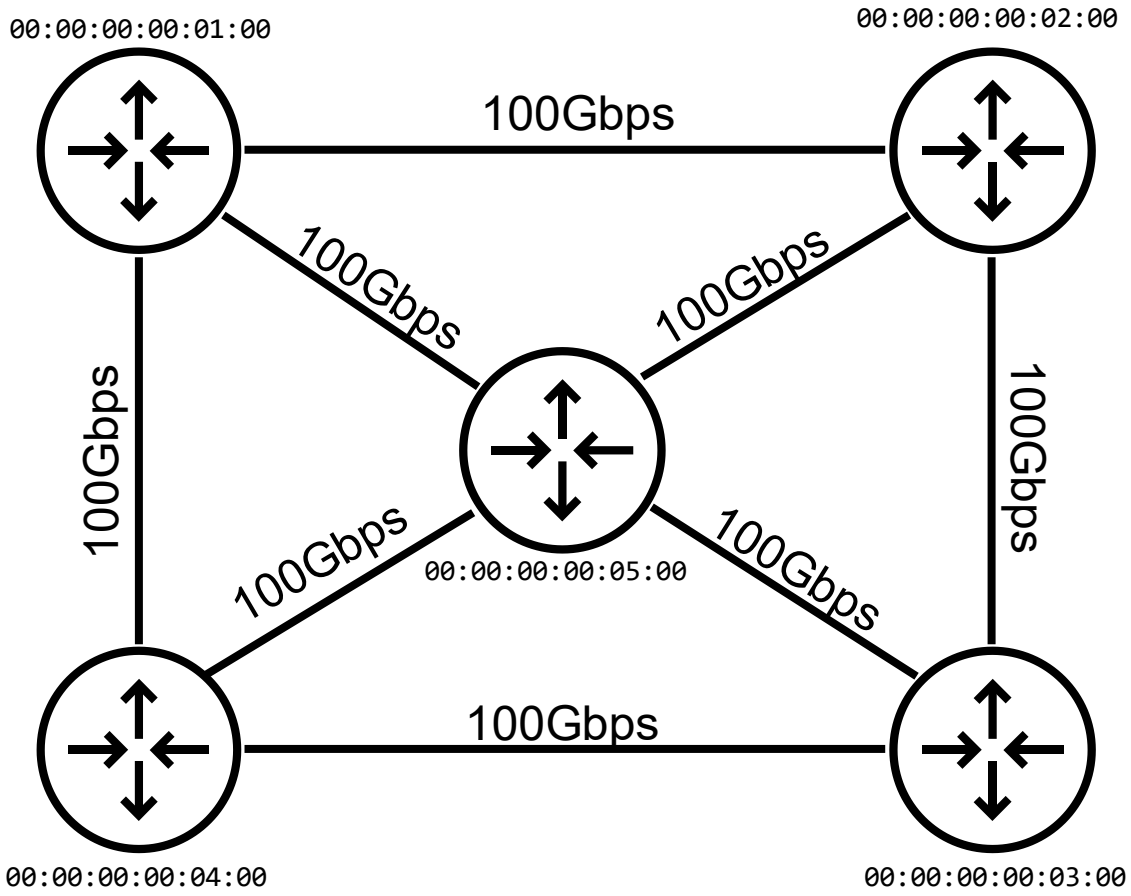
- For example, Voice-over-IP (phone calls) and data traffic
- Different services may use different paths in the network

SPB offers the possibility of introducing services

- Splitting the network into multiple logical networks
- Efficient distribution of multicast and broadcast traffic
- Avoiding bottlenecks due to conflicting services

Each service is identified by a service ID (I-SID)

- If multiple trees of equal costs are available, each services can use different trees
- Thus, the tree used is identified based on the service and edge switch



Services in SPB

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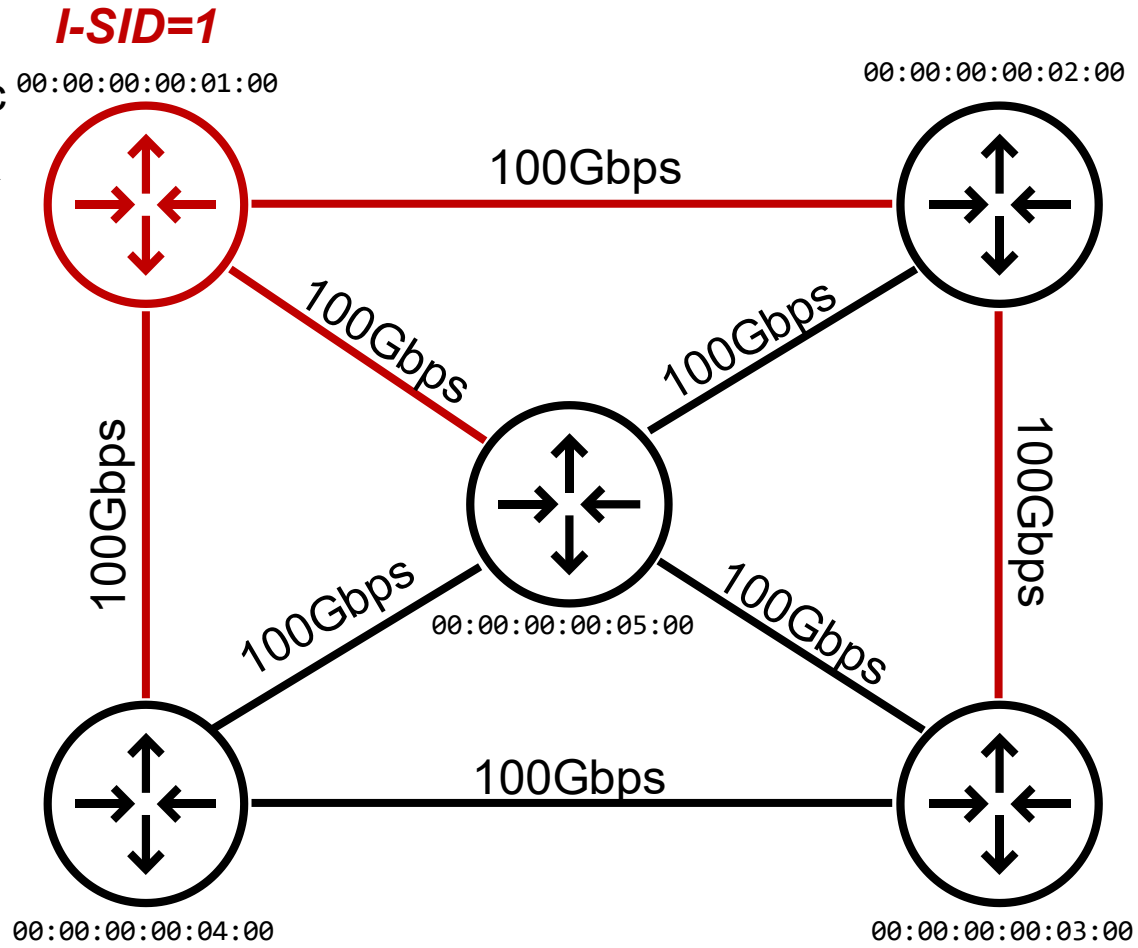
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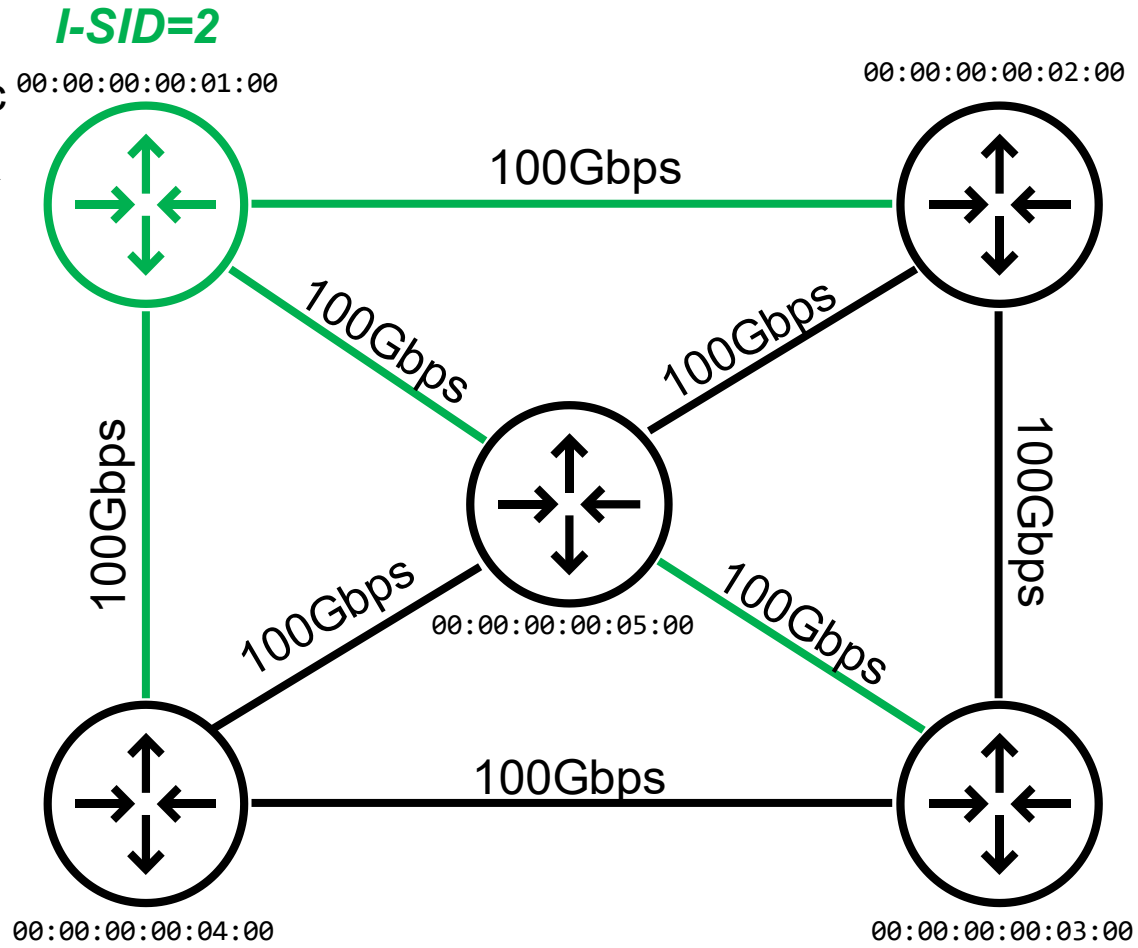
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- For example, Voice-over-IP (phone calls) and data traffic
- Different services may use different paths in the network

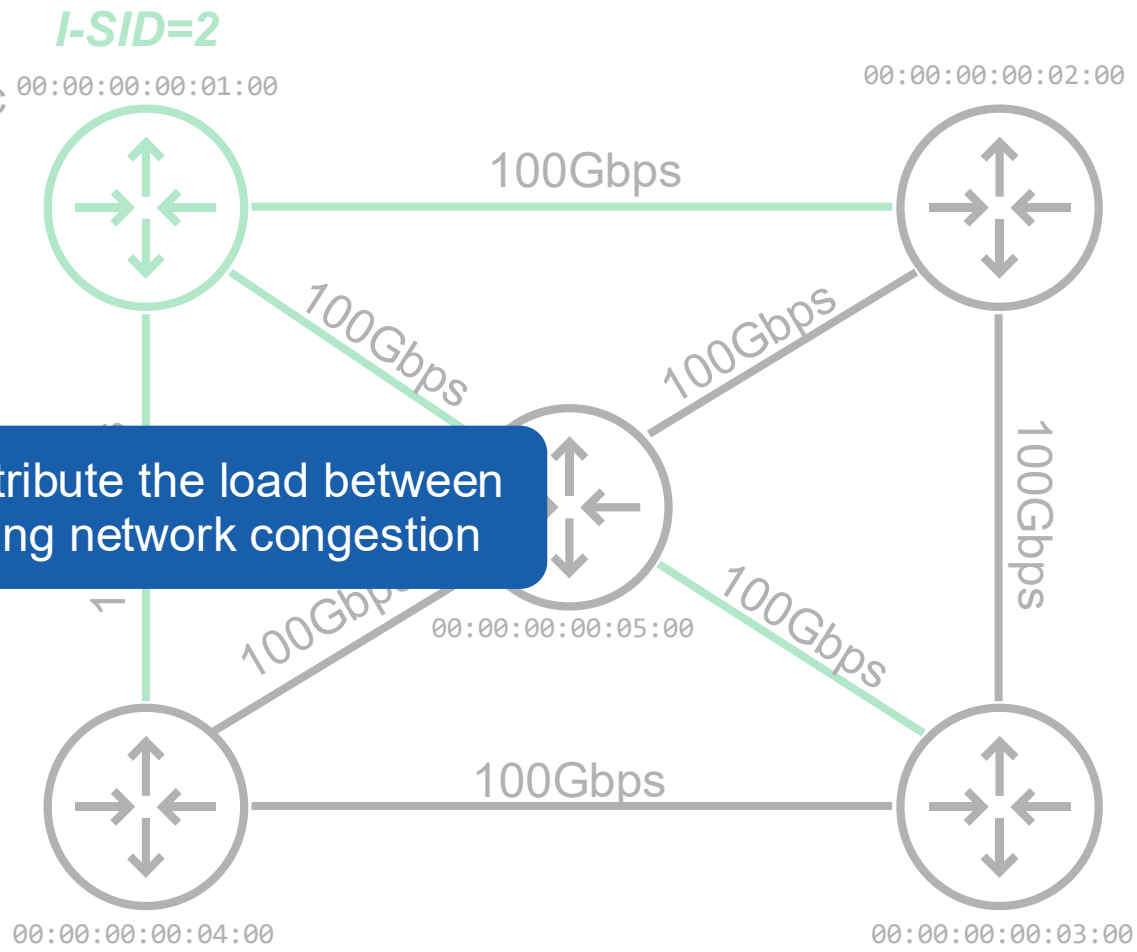
SPB offers the possibility of introducing services

- Splitting the network into multiple logical networks
- Efficient distribution of
- Avoiding bottlenecks c

Each service is identified by a service ID (I-SID)

- If multiple trees of equal costs are available, each services can use different trees
- Thus, the tree used is identified based on the service and edge switch

Different shortest path trees distribute the load between paths of equal cost, thus avoiding network congestion

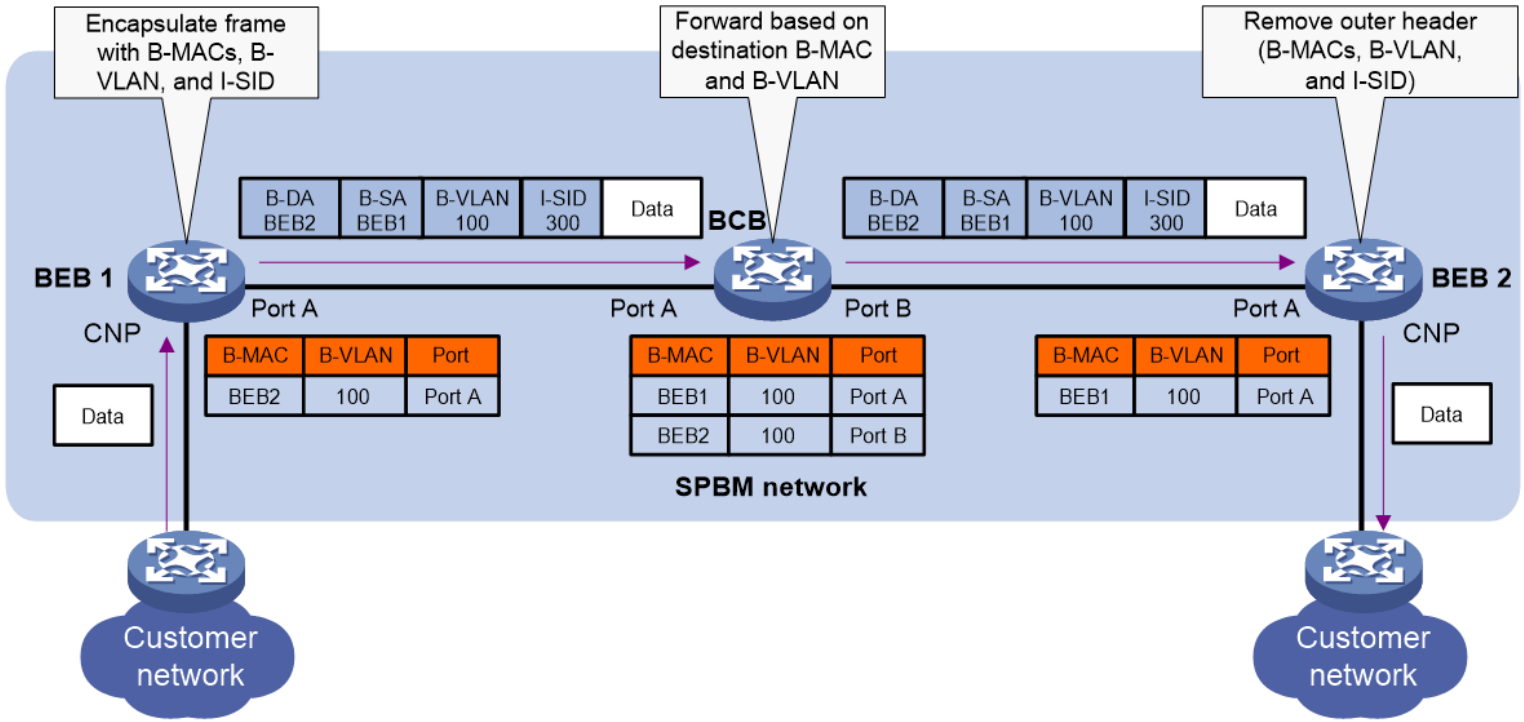


Encapsulation in SPBM

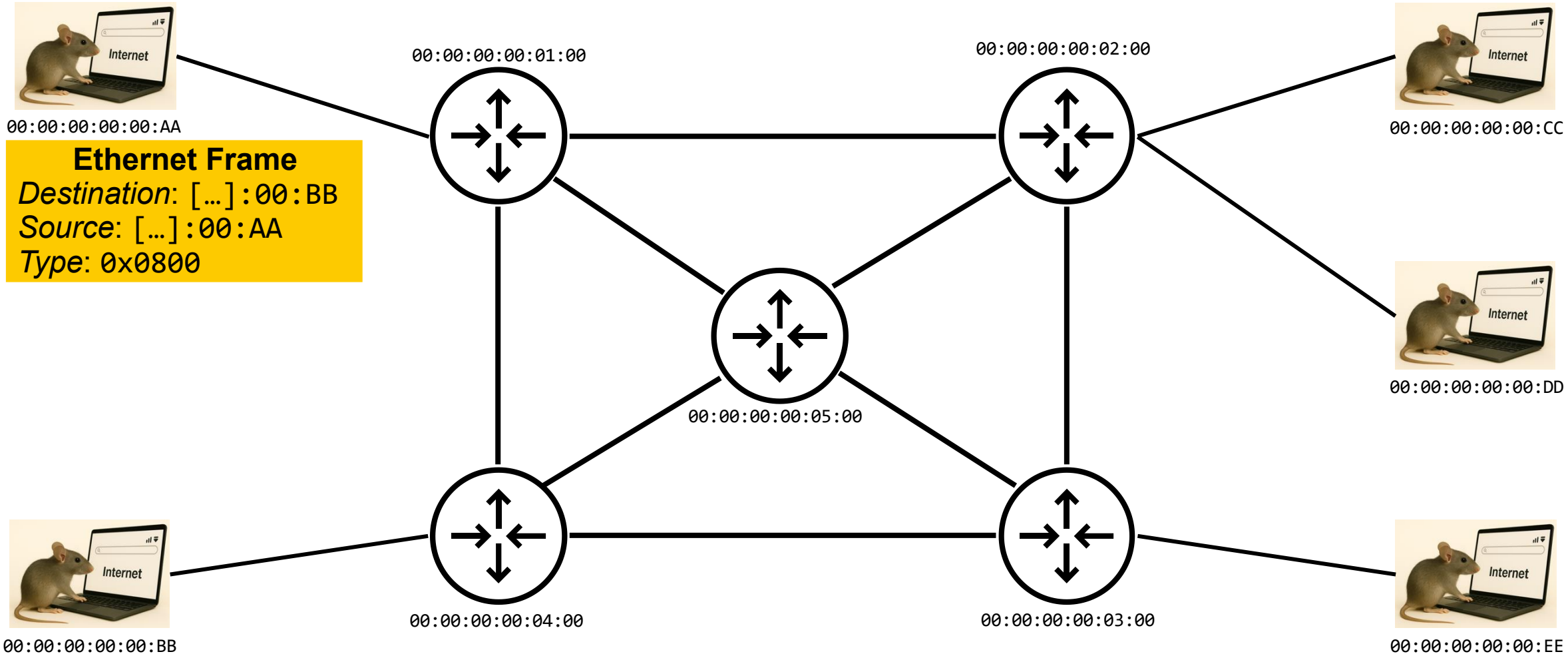
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SPBM encapsulates frames in another frame once it first arrives at a bridge

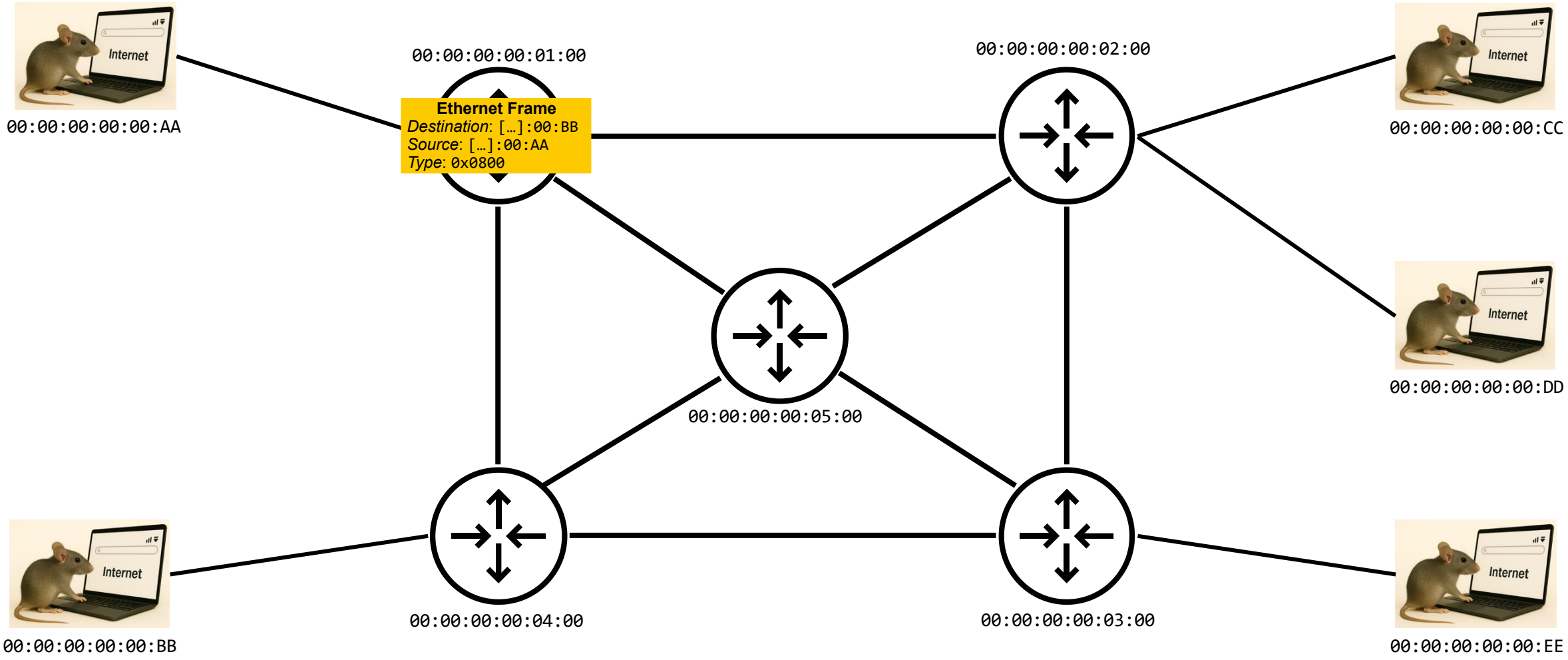
- Encapsulating frame has the source MAC of the bridge and the target MAC of the last bridge on the path
- The last bridge then removes the header and forwards the packet to the destination
- For multicast/broadcast, the destination is set to the respective multicast/broadcast address



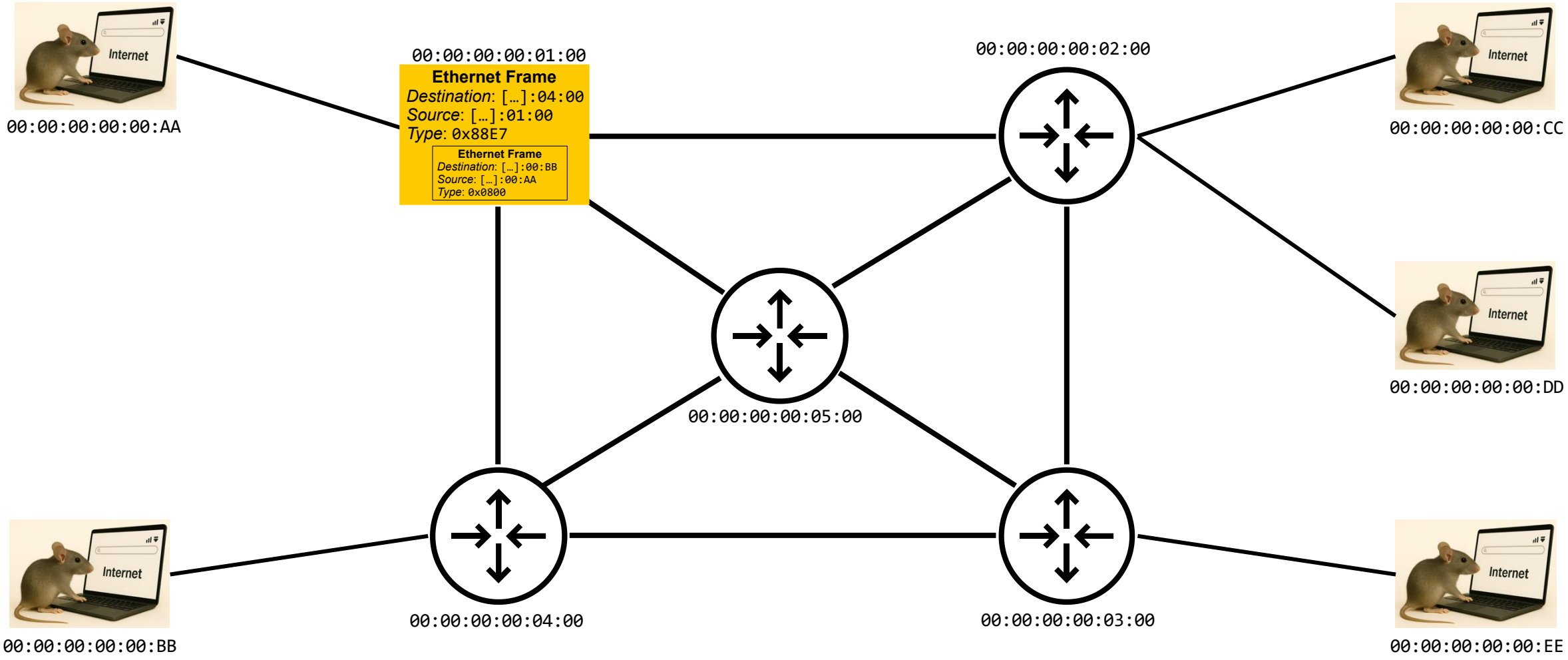
Unicast Routing in SPBM



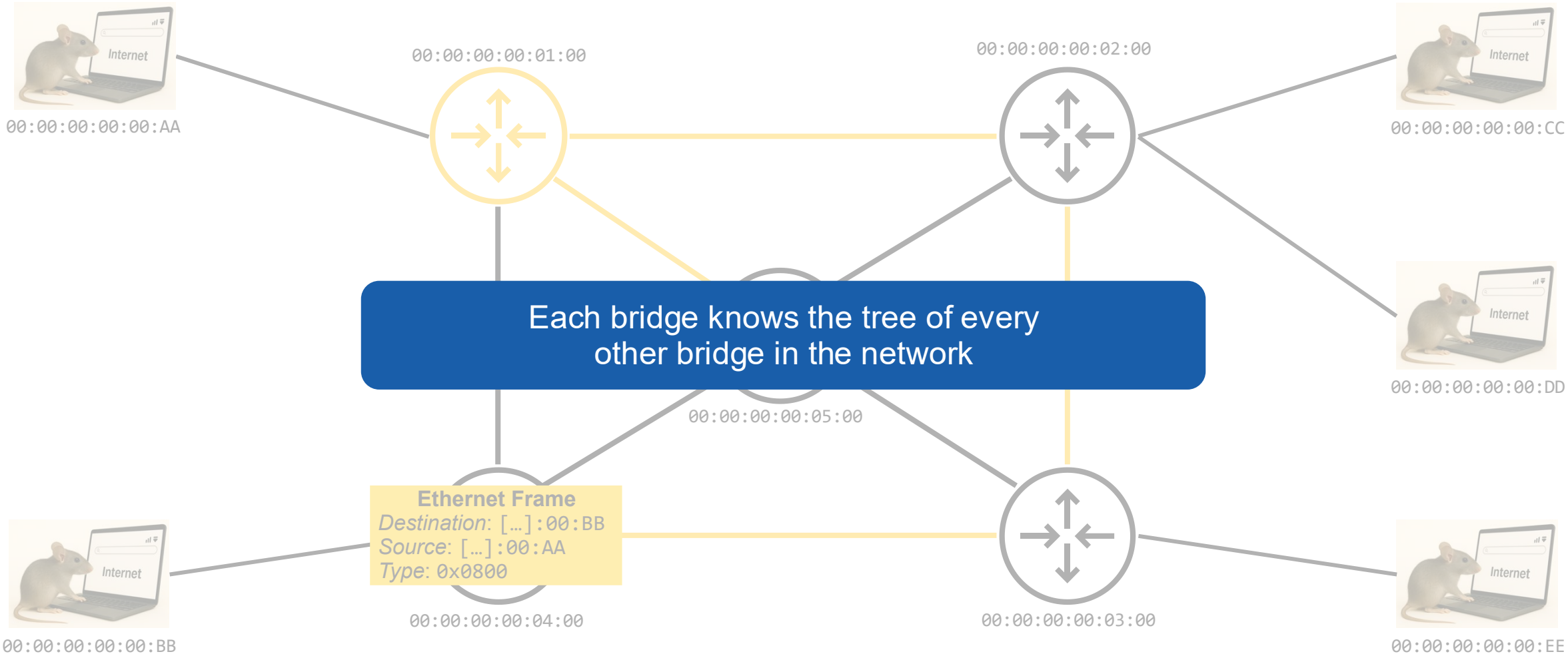
Unicast Routing in SPBM



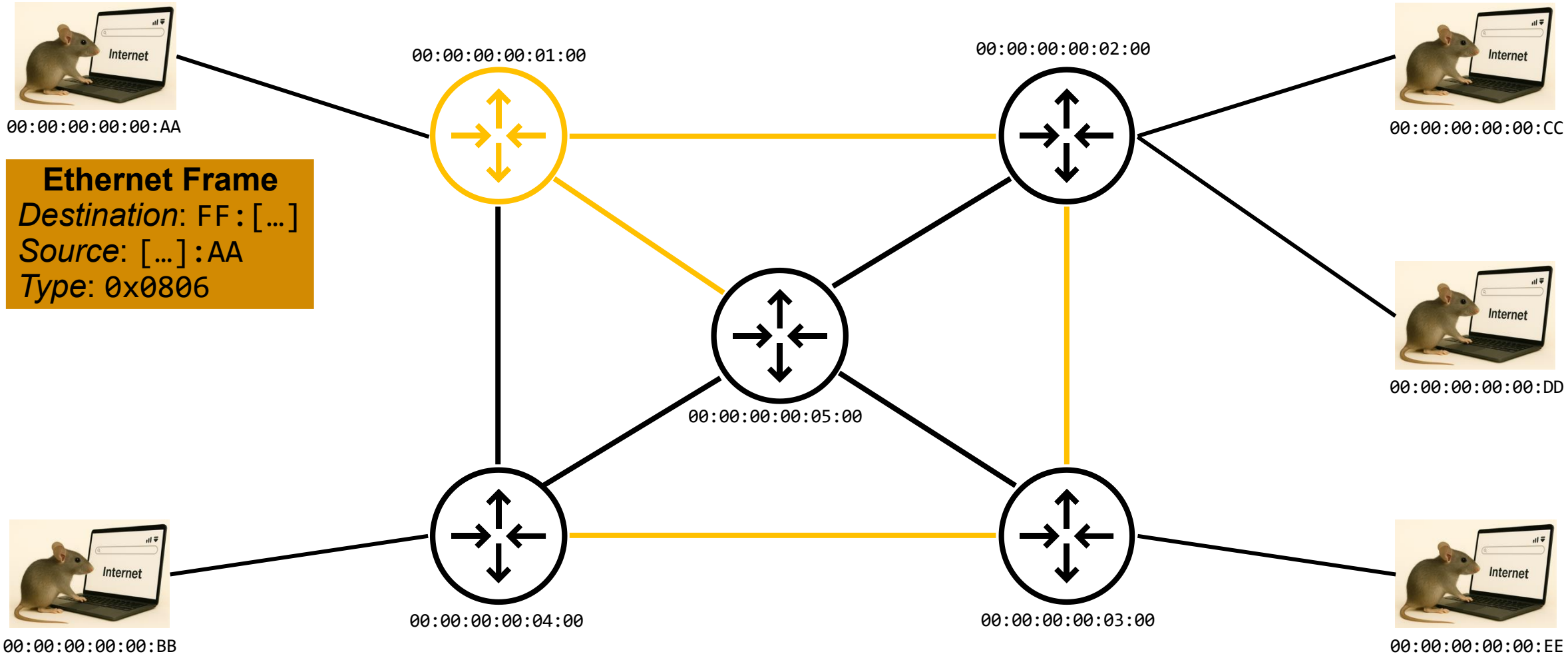
Unicast Routing in SPBM



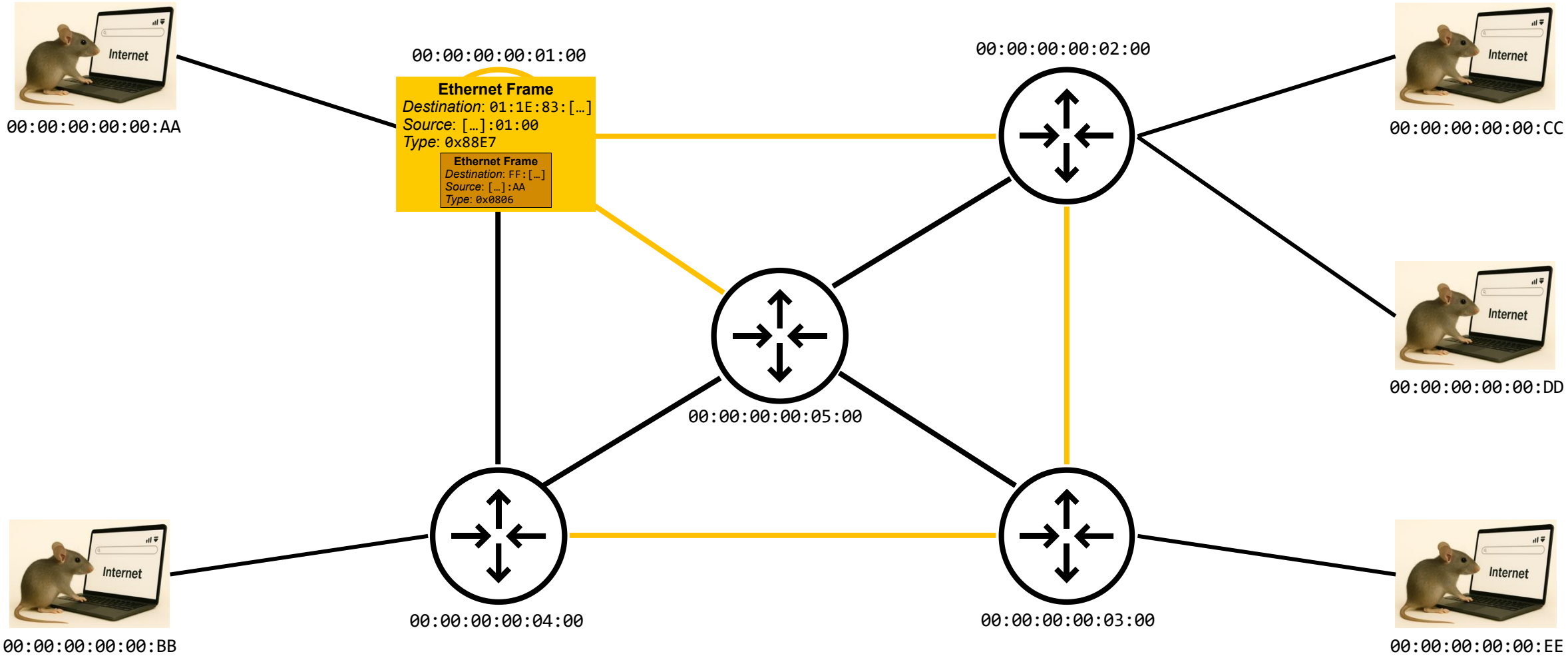
Unicast Routing in SPBM



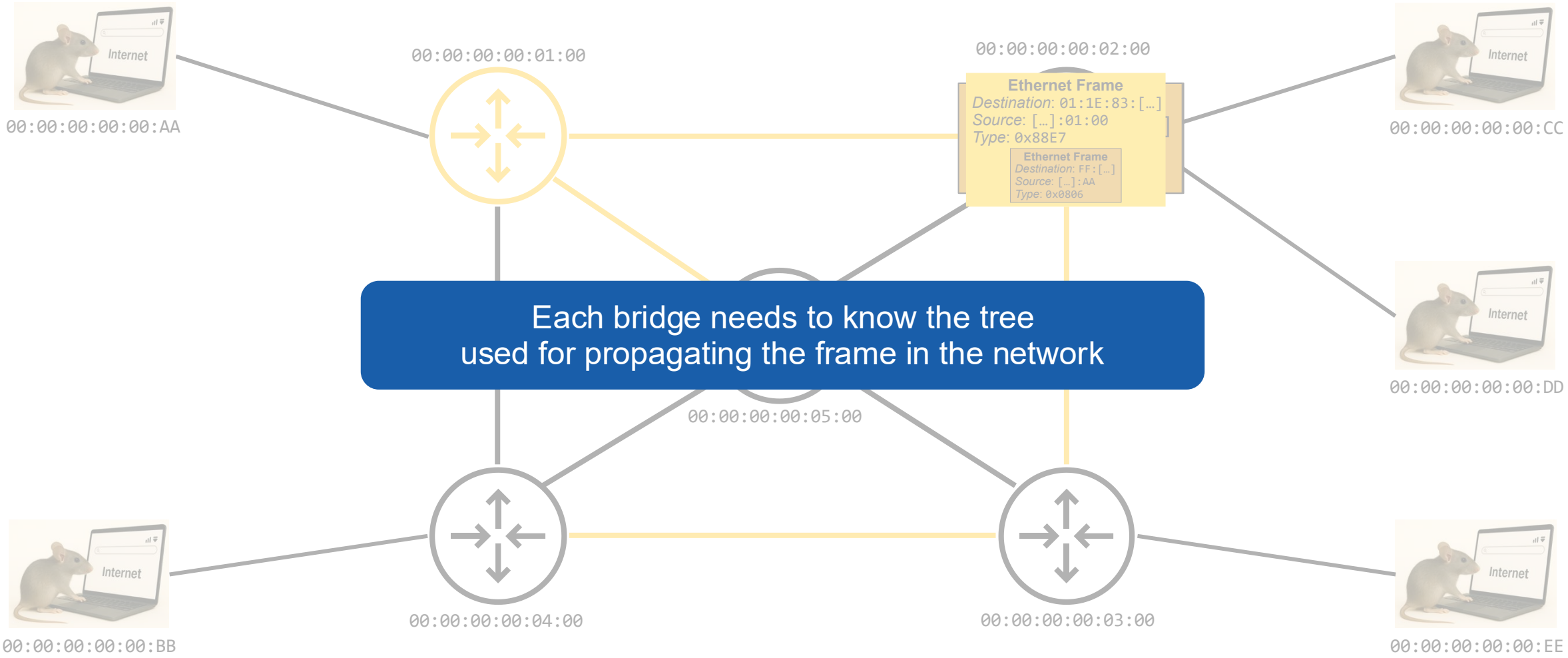
Broadcast Routing in SPBM



Broadcast Routing in SPBM



Broadcast Routing in SPBM



Failure Recovery of SPB

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In contrast to STP, SPB utilizes knowledge about the full topology

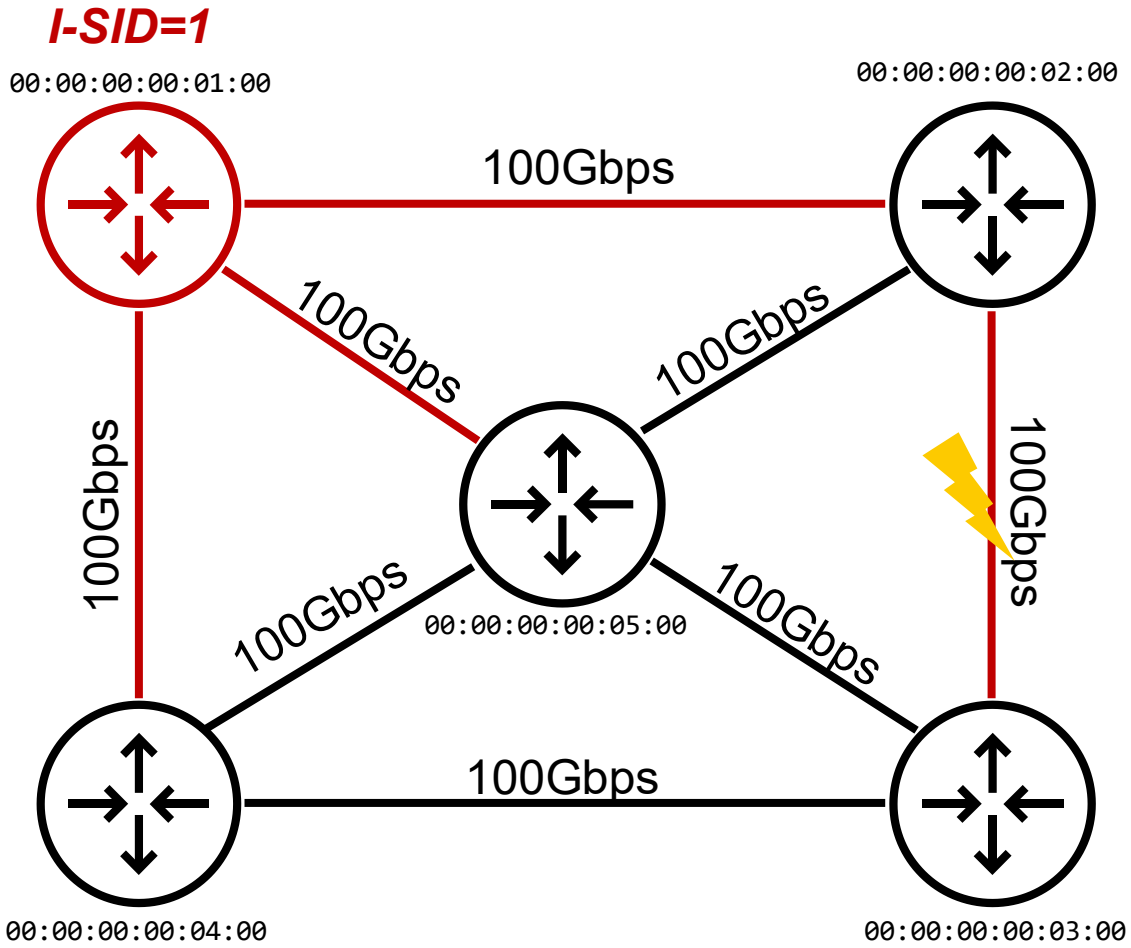
- Convergence is almost instantaneous, as each switch updates recalculates its tree once it is aware of a link or switch failure
- The new trees are shared with all other switches in the network

How to detect link failures?

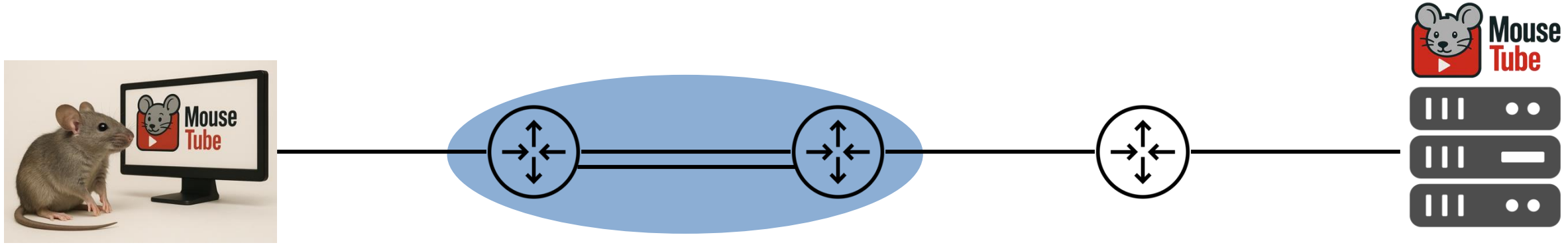
Failure detection

- Direct neighbors use Hello-packets to constantly check the link
- If a link is down, a link-state advertisement (LSP) is shared with all switches, signaling that the link is down

How to detect node failures?



… but we want path redundancy and *fast* failover



Hide the redundancy with logical links

Use alternatives to (R)STP

Link Aggregation

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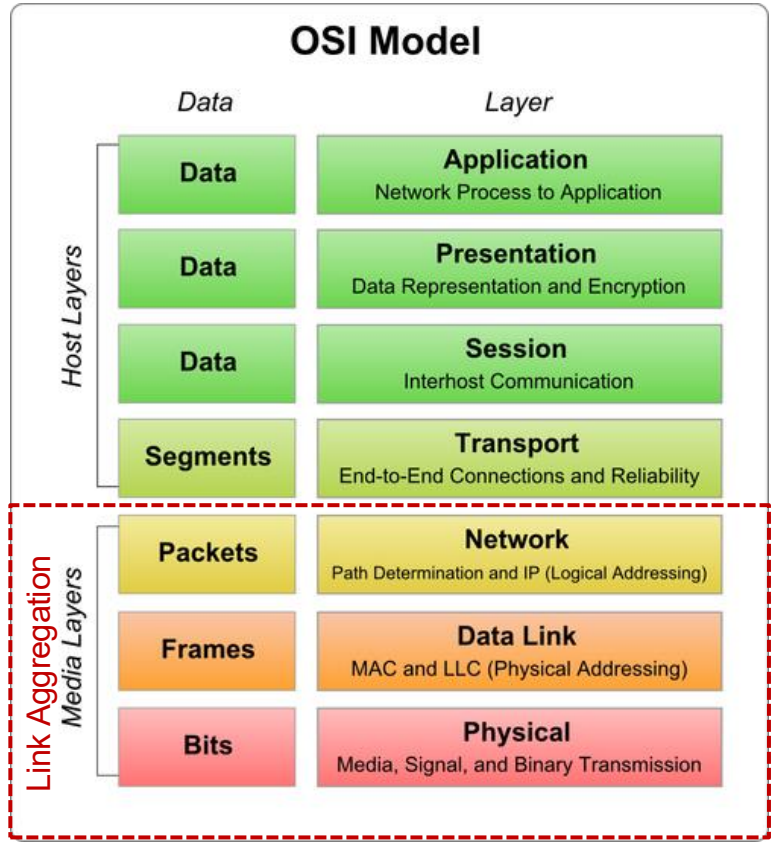
Requirement: Two switches/routers are connected via multiple fibers

- Increase robustness against fiber cuts
- Increase throughput between the two switches/routers

General idea: Combine both physical links to one virtual link

For example

- Send packets via both links simultaneously (higher robustness, but similar throughput)
- Send packets via either link (only slightly higher robustness, but increased throughput)



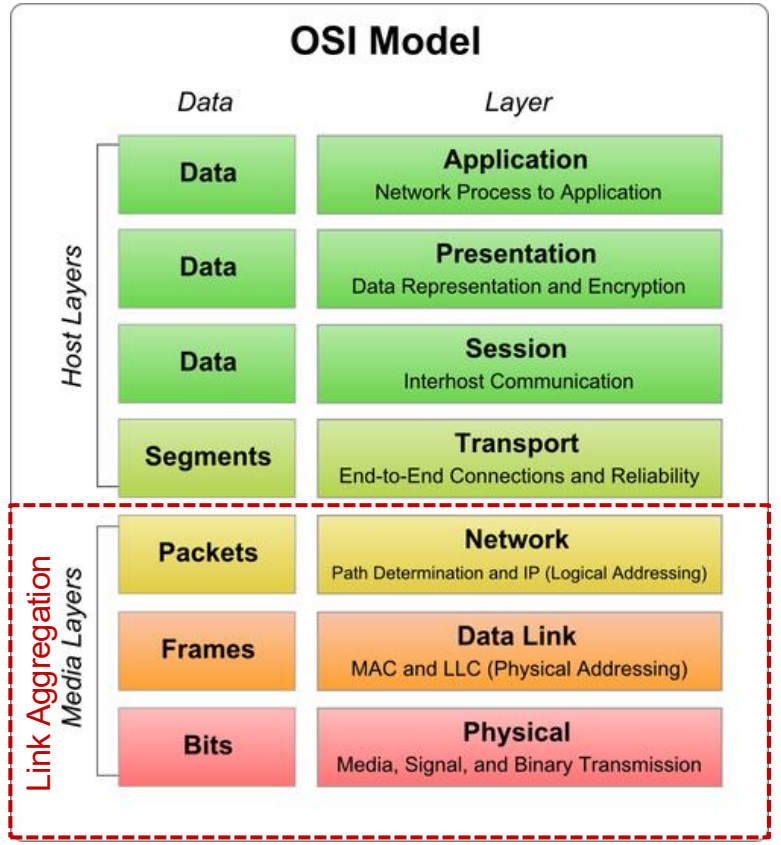
Link Aggregation – General information

The collection of ports is called Link Aggregation Group (LAG)

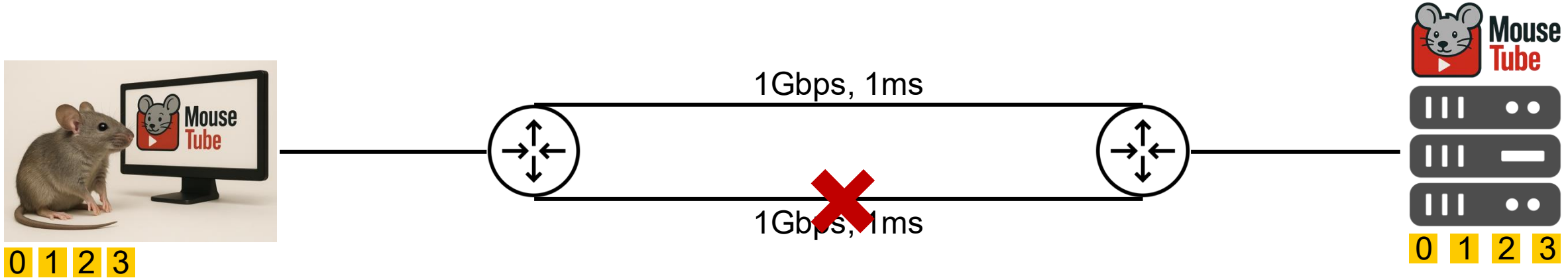
- In theory, each port except one in a LAG can fail without loosing connectivity
- Addresses the port at each switch/router, the fiber/cable, and the transceiver as single-point-of-failure

Examples include

- Devices using multiple frequency bands on the physical layer
- Combining multiple ethernet ports to one logical port
- Transmission via multiple IP-based interfaces



Simplified scenario

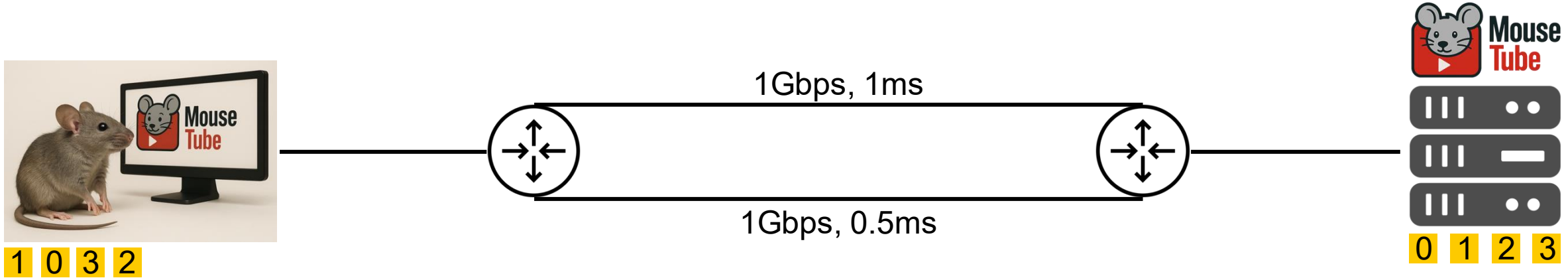


Transmission of packets in round-robin fashion over two interfaces

- Both links are utilized, and the effective bandwidth is combined
- All packets arrive correctly at the receiver
- Even if one link fails, the packet can be forwarded via the other link

Sounds simple. Why should we talk about that here?

Simplified scenario



MouseTube controls two switches on the path and wants to utilize link aggregation

- What happens to packets send over alternating links?

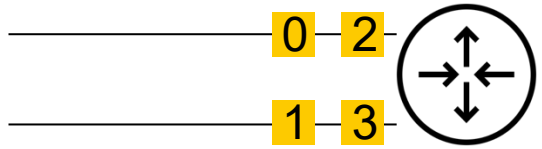
Packet ordering is destroyed! Issues for applications and congestion control

Packet Scheduling in Link Aggregation

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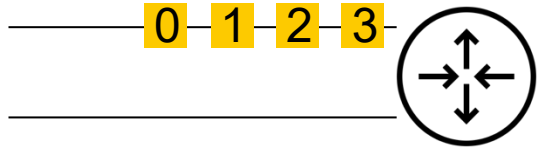
Round-robin-based

- Transmit each packet via a different link in the order of links in the LAG
- Does not ensure in-order delivery of packets



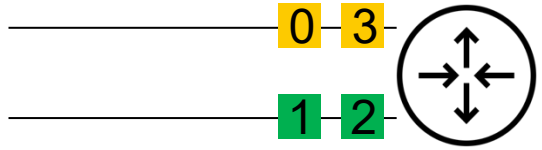
Active-Backup

- Only use one link in the LAG, only switch to another link if the main link is down
- Only provides fault tolerance, no performance increase



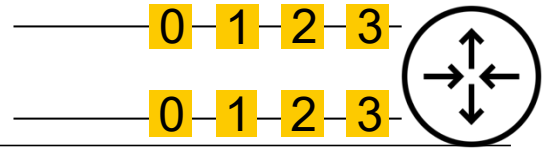
Hash-based scheduling

- Use a hash considering source and destination address and port to determine the utilized link
- Ensure that packets belonging to the same connection are sent via the same link
- Most frequently used to benefit from both fault tolerance and increased throughput



Broadcast

- Transmit packets via all links simultaneously



Hash-based scheduling: CRC-based

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Cyclic redundancy check (CRC) is an error-detecting code

- Produces a “hash” of the input data
- Very simple algorithm without cryptographic safety, but easy to compute
- Easy to implement in hardware
- Normally used to detect errors caused by noise in transmission channels
- Utilizes a so-called generator polynomial of length *n*
- This polynomial needs to be know by both communication partners

Basic idea: Use the hash based on flow properties to assign packets to a link

The hash output of CRC is a binary number of fixed length

- Can be used to assign packets to a link by calculating the remainder of the hash given the number of links

CRC-16	0x8005	$x^{16} + x^{15} + x^2 + 1$
CRC-CCITT	0x1021	$x^{16} + x^{12} + x^5 + 1$
CRC-DNP	0x3D65	$x^{16} + x^{13} + x^{12} + x^{11} + x^{10} + x^8 + x^6 + x^5 + x^2 + 1$
CRC-32	0x04C11DB7	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x^1 + 1$

Hash-based scheduling: CRC-based
Example of 3-bit CRC

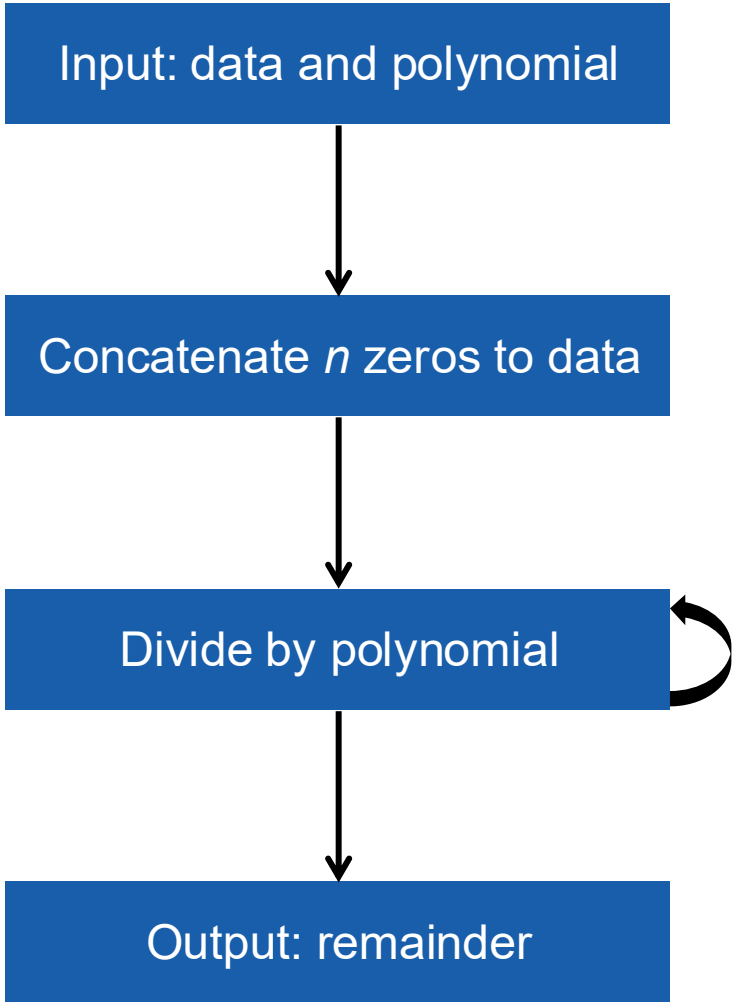
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Input

- Data: `0100100` (1 byte)
- Polynomial `1010` (3-bit CRC)

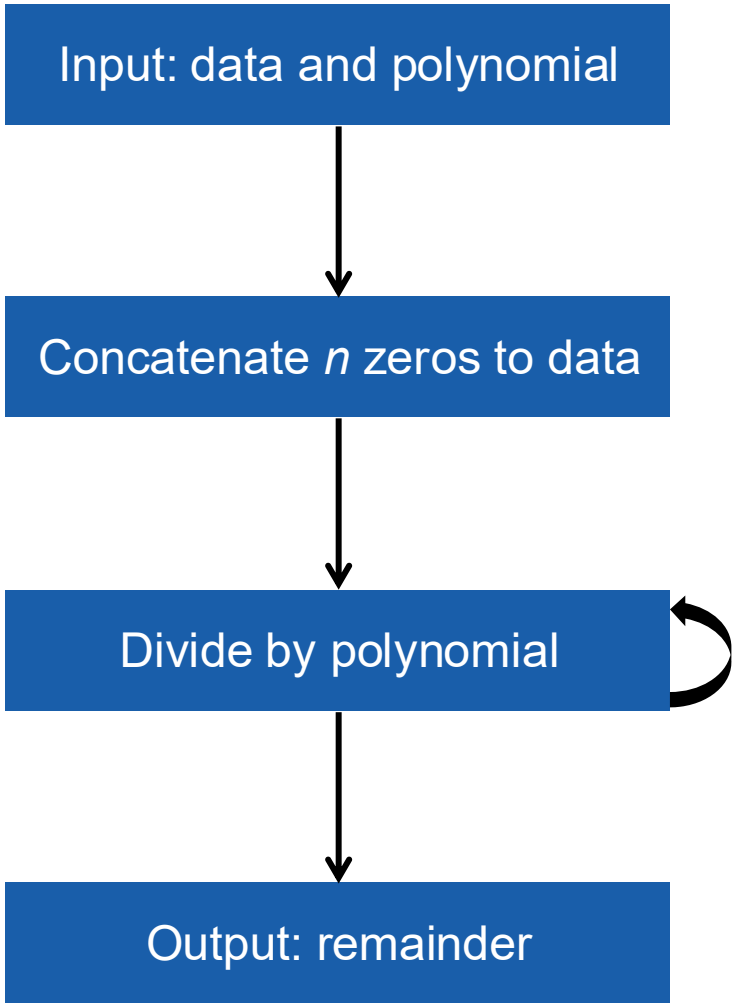
Padded

- `0100100 000`



Hash-based scheduling: CRC-based
Example of 3-bit CRC

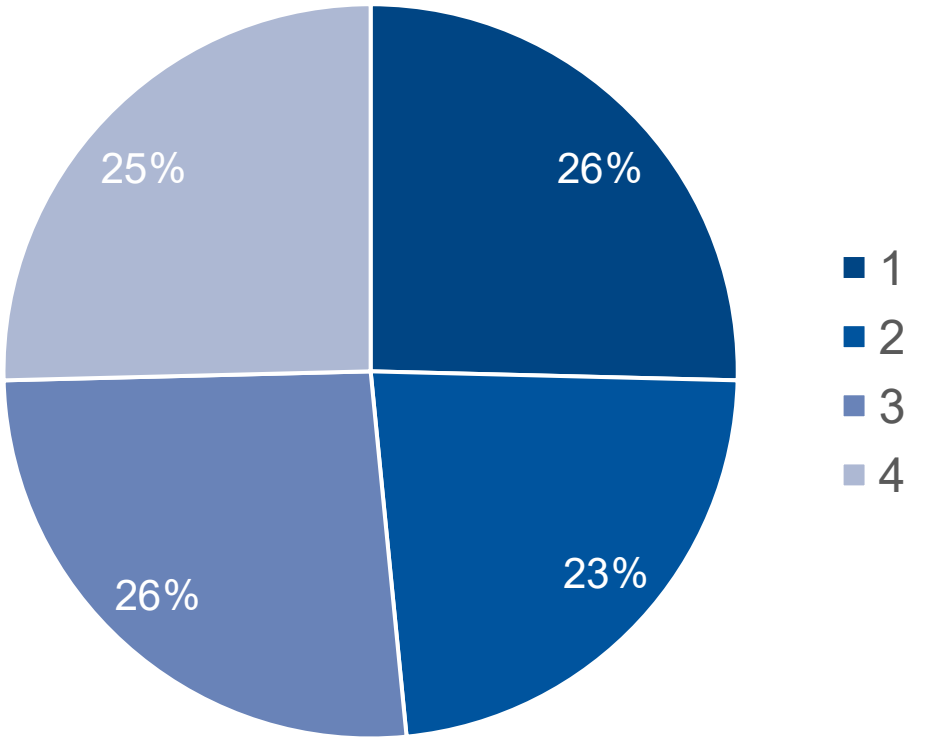
```
0100100 000
 1010
-----------
0001100 000
   1010
-----------
0000110 000
    101 0
-----------
0000011 000
     10 10
-----------
0000001 100
      1 010
-----------
      0 110 --> remainder
```



Examples for a LAG with CRC-32 and 4 links

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IP address	CRC-32	Link #
10.0.30.0	0x56b96cbc	0
10.0.30.1	0x1083a946	2
10.0.30.2	0x4dad8881	1
10.0.30.3	0x48f0f1b3	3
10.0.30.4	0x8022fd4a	2
10.0.30.5	0x448b65fe	2
10.0.30.6	0x8843bbf3	3
10.0.30.7	0xf63e0f90	0
10.0.30.8	0x22fb42c6	2
10.0.30.9	0xa027734a	2
10.0.30.10	0x44ab6070	0
10.0.30.11	0x824335d1	1
10.0.30.12	0x23544efa	2
10.0.30.13	0x8f49a0c1	1
10.0.30.14	0x3eee540b	3
10.0.30.15	0x45eed851	1



Link Aggregation Control Protocol (LACP)

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Common and vendor-independent standard for Ethernet

- Specified in IEEE 802.1AX
- Method of bundling multiple physical links into one logical link

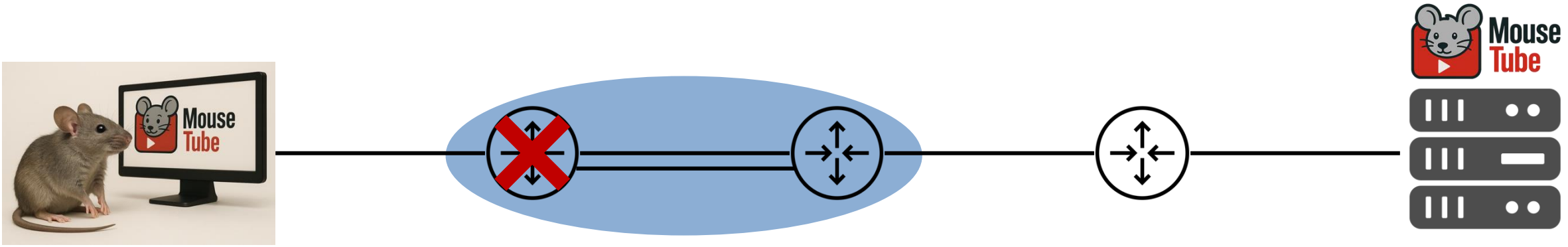
Negotiation with peer devices to establish and maintain a LAG

- Configuration of the number of bundled links
- Configure packet scheduling (if applicable)
- Detect failures of individual links

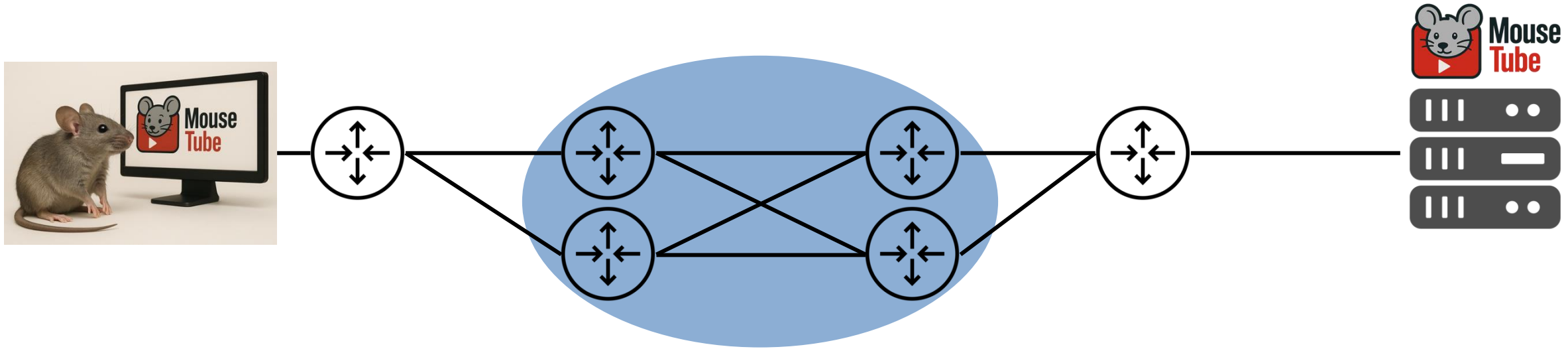
Limitations

- Requires the same links speed of all ports
- Protects against failures of links, but not against node failures

What would link redundancy mean for MouseTube?



What would link redundancy mean for MouseTube?

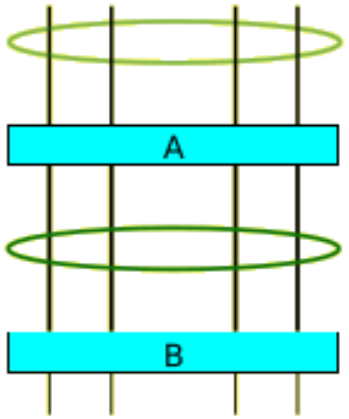


Multi-chassis link aggregation group (MLAG)

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Extension to LAG to be resilient to node failures

- Instead of utilizing only multiple links, also multiple chassis (devices) are used
- Coordination between devices is implementation-dependent and varies between vendors
- Coordinated chassis act like a single physical switch



(1) LAG

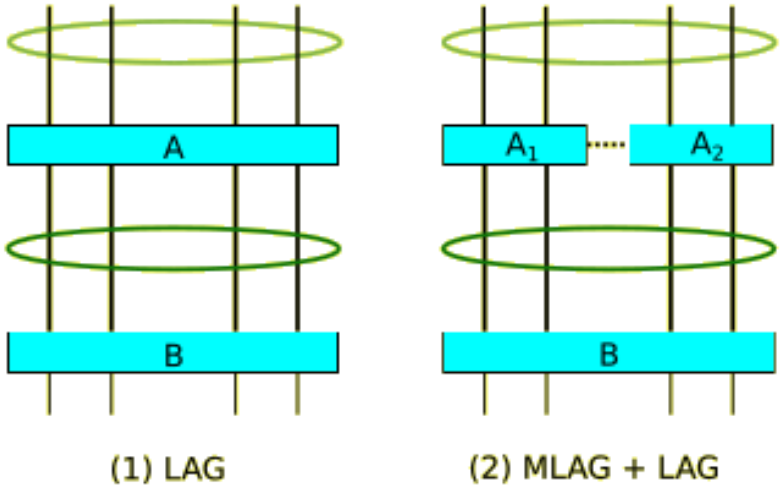
https://en.wikipedia.org/wiki/Multi-chassis_link_aggregation_group#/media/File:Using_Multi-Chassis_LAG_(MC-LAG)_for_High_Availability.png

Multi-chassis link aggregation group (MLAG)

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Extension to LAG to be resilient to node failures

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(1) LAG (2) MLAG + LAG

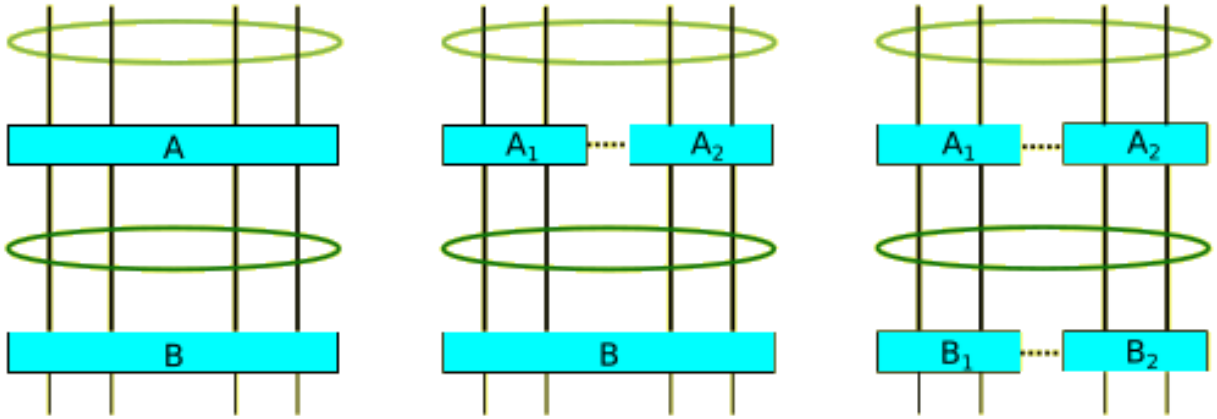
https://en.wikipedia.org/wiki/Multi-chassis_link_aggregation_group#/media/File:Using_Multi-Chassis_LAG_(MC-LAG)_for_High_Availability.png

Multi-chassis link aggregation group (MLAG)

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Extension to LAG to be resilient to node failures

- Instead of utilizing only multiple links, also multiple chassis (devices) are used
- Coordination between devices is implementation-dependent and varies between vendors
- Coordinated chassis act like a single physical switch



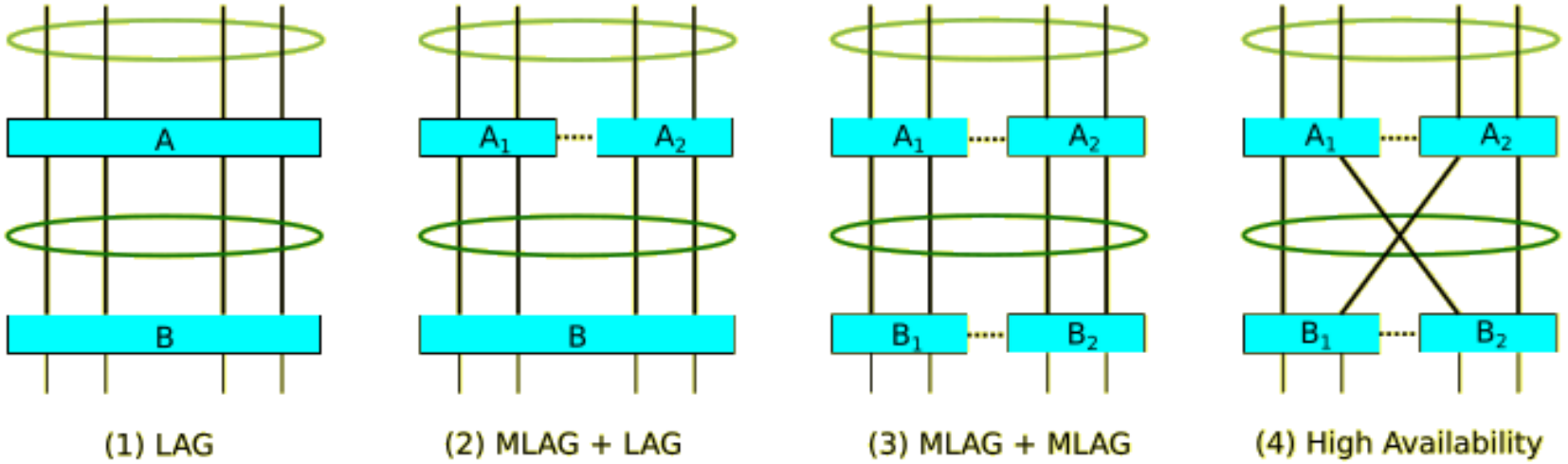
(1) LAG (2) MLAG + LAG (3) MLAG + MLAG

Multi-chassis link aggregation group (MLAG)

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Extension to LAG to be resilient to node failures

- Instead of utilizing only multiple links, also multiple chassis (devices) are used
- Coordination between devices is implementation-dependent and varies between vendors
- Coordinated chassis act like a single physical switch



(1) LAG (2) MLAG + LAG (3) MLAG + MLAG (4) High Availability

https://en.wikipedia.org/wiki/Multi-chassis_link_aggregation_group#/media/File:Using_Multi-Chassis_LAG_(MC-LAG)_for_High_Availability.png

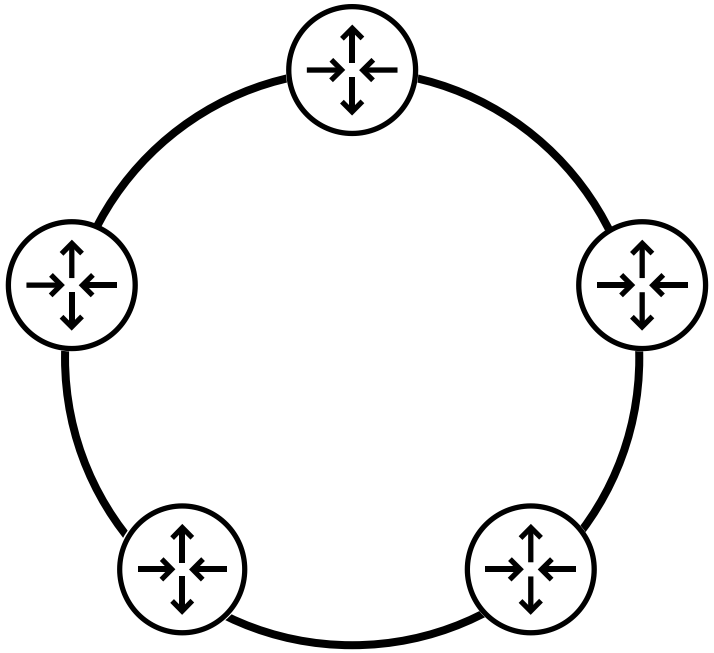
Efficiency of Ethernet links

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All the networks we considered thus far had a large number of excess links

- Works well in a local area network, in which the links are relatively short
- Works quite poor in metropolitan area networks, in which links are quite long

What is the topology with the lowest number of links that offers basic resilience?



Ethernet Ring Protection Switching (ERPS)

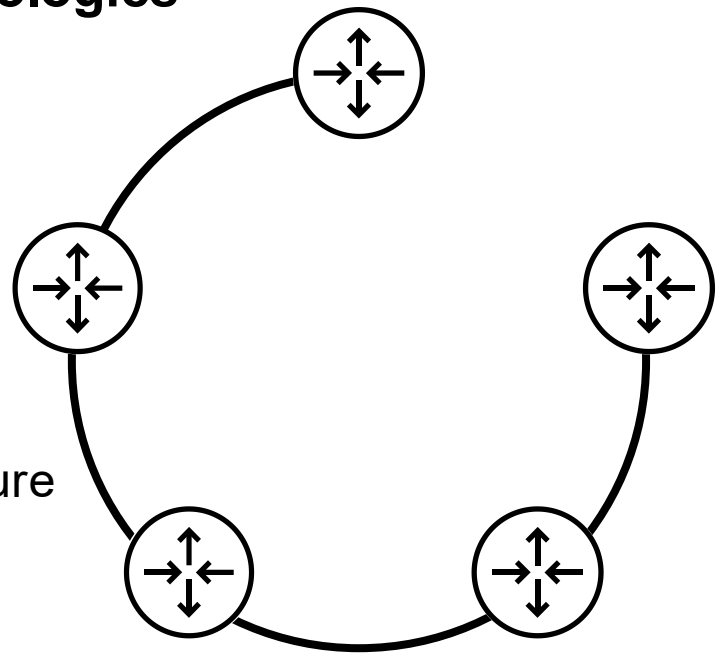
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Shortest path bridging (SPB) or Spanning Tree Protocol (STP) are not required in this case

How do you prevent a loop in a ring topology?

Ethernet Ring Protection Switching enables switching in ring topologies

- ITU-T recommendation under G.8032
- Version 1 supports a single ring topology, while Version 2 supports multi ring / ladder topologies



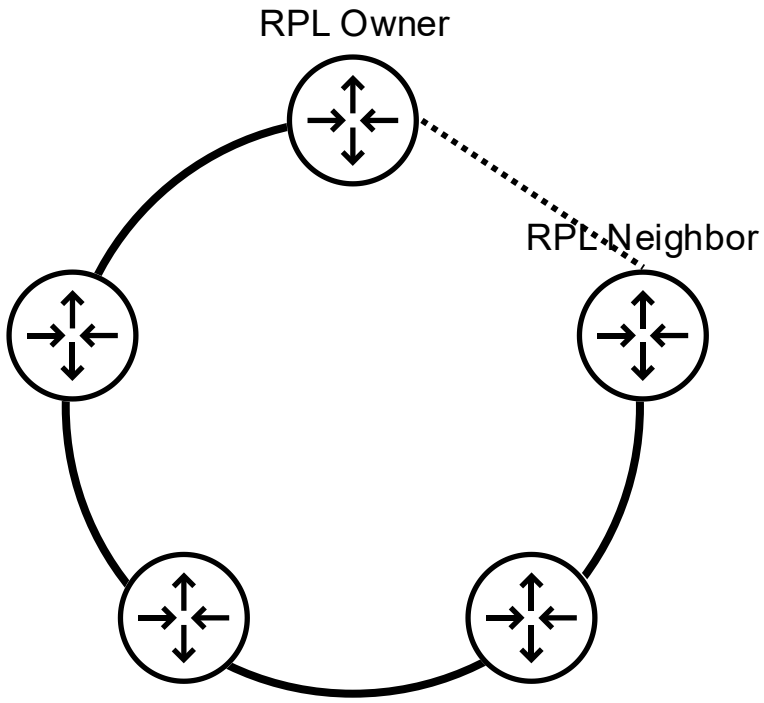
For loop avoidance, ERPS disables exactly one link in the ring

- The disabled link is called Ring Protection Link (RPL)
- The link is blocked during normal operation, and only enabled during a failure
- The RPL Owner is responsible for blocking/unblocking the RPL
- The other end of the RPL may participate in blocking, but is not required

ERPS: Exchanged Messages

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Ring Automatic Protection Switching (R-APS) Messages are exchanged to coordinate the state of the RPL

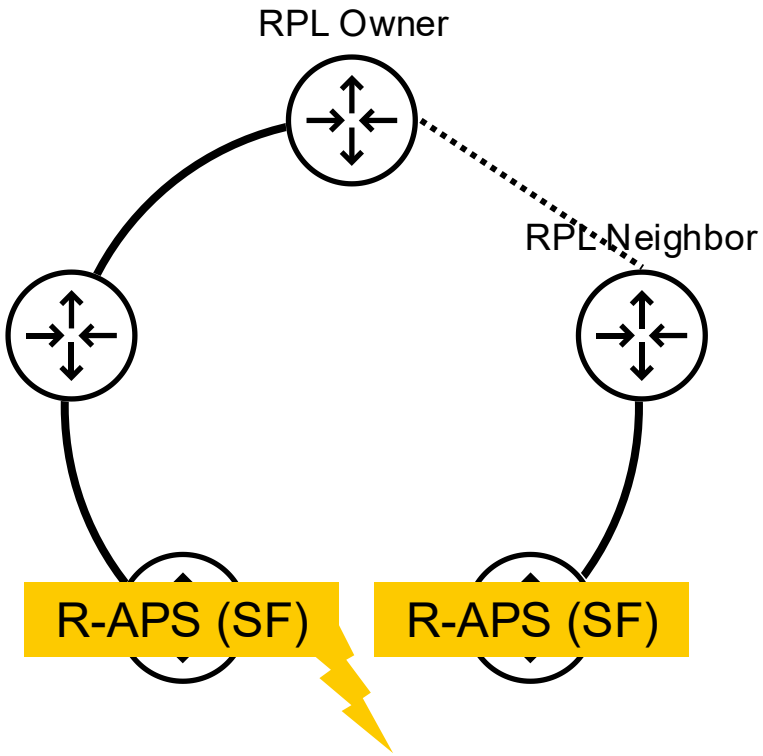


ERPS: Exchanged Messages

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Ring Automatic Protection Switching (R-APS) Messages are exchanged to coordinate the state of the RPL

- R-APS (SF): Signal failure
  - Transmitted by the neighbor of a failed link after the respective port has been blocked
  - Used by the RPL Owner and RPL Neighbor to unblock the RPL

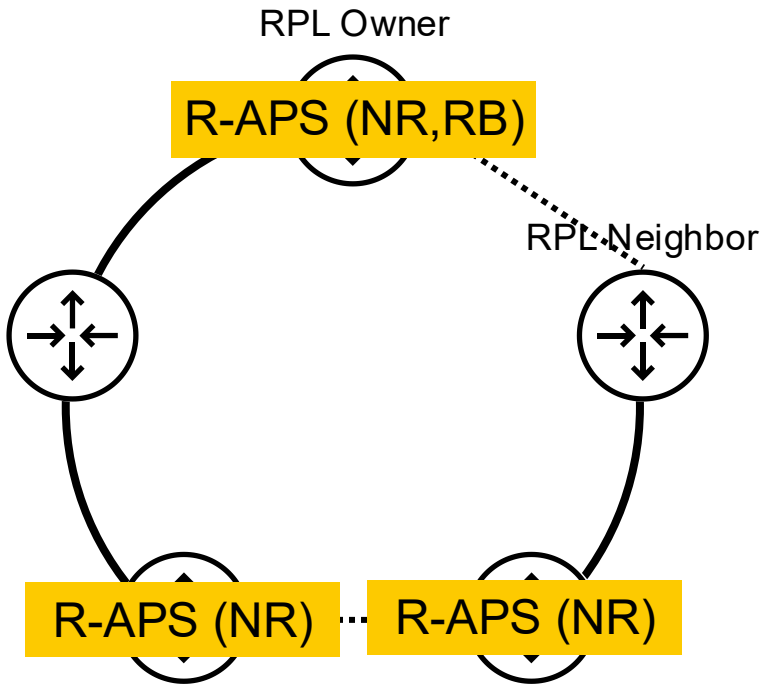


ERPS: Exchanged Messages

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Ring Automatic Protection Switching (R-APS) Messages are exchanged to coordinate the state of the RPL

- R-APS (SF): Signal failure
  - Transmitted by the neighbor of a failed link after the respective port has been blocked
  - Used by the RPL Owner and RPL Neighbor to unblock the RPL
- Once the link has recovered: R-APS (NR): No request
  - Transmitted by the neighbor of a recovered link
  - On receiving the message, the RPL Owner blocks the RPL
- R-APS (NR, RB): No request, RPL blocked
  - Transmitted by the RPL Owner
  - On receiving the message, all other nodes unblock all connected links



ERPS v2: Multi Ring Support

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In Version 2, ERPS also supports multi ring topologies

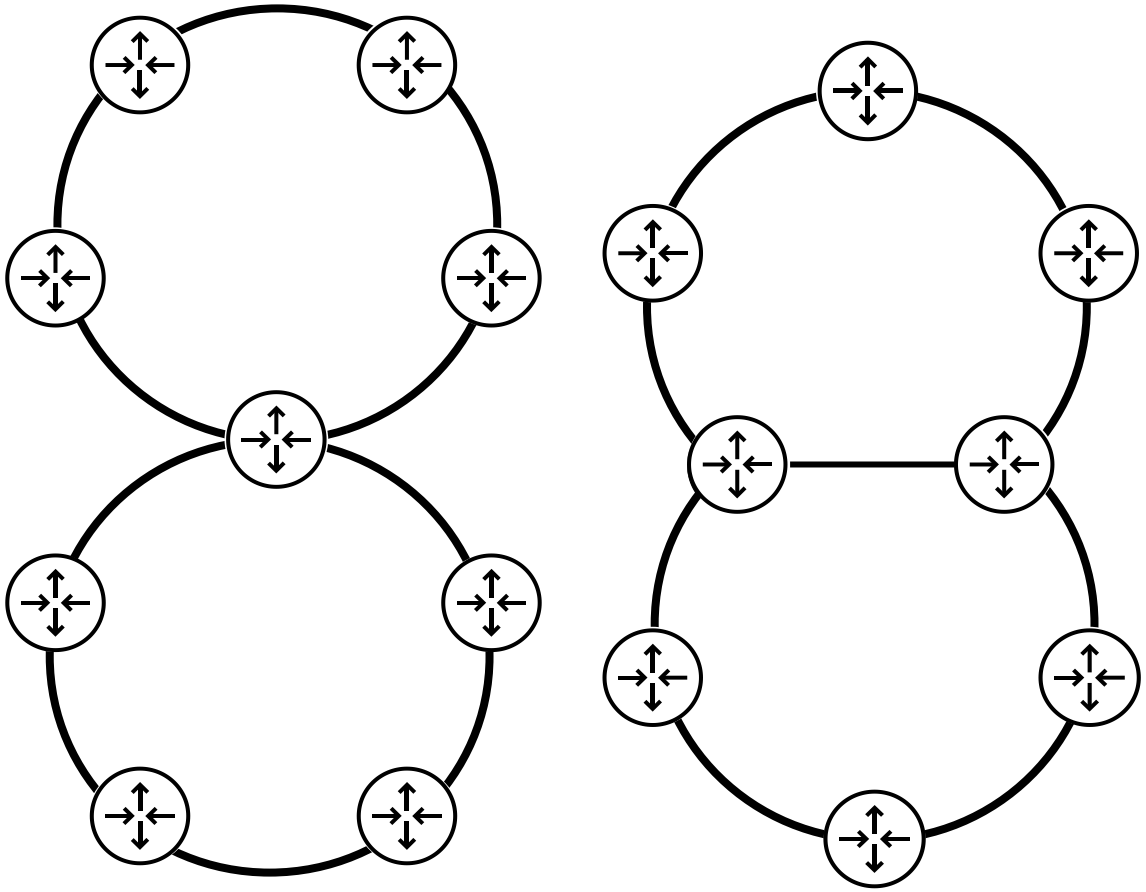
What do you think was the issue with just applying v1?

Version 1 did not introduce any separation between multiple rings

- R-APS messages of one ring would be associated also with the other ring
- Ultimately leading to unstable behavior

Version 2 separates the rings using VLAN

- Each ring has its own VLAN to identify its own messages
- Nodes which are part of two rings process messages of both rings, but forward them only accordingly



What have we learned today?

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Link Aggregation Groups (LAGs) can reduce the risk of link failure

- Parallel utilization of multiple parallel links between two or more devices
- MLAG can expand link aggregation to multiple switches

Shortest Path Bridging (SPB) can already achieve much lower recovery times compared to STP

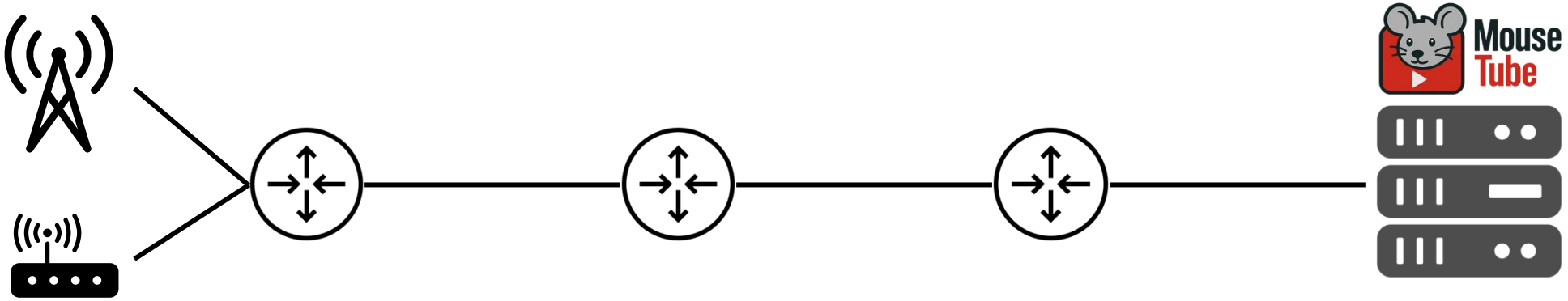
- Fast convergence due to the utilization of full topology information
- Better utilization of network resources

Ethernet Ring Protection Switching (ERPS)

- Provides basic levels of resilience of ring-based Ethernet networks
- Is commonly applied if the creation of new physical links is very expensive

Side Note: Wireless Resilience





Often, consumer devices use wireless access to connect to the Internet

- Wifi
- Cellular communication (5G, …)

Wireless communication is more challenging

- Transmission errors are more common
- Concurrent transmissions may affect each other

What resilience mechanisms are employed in wireless communication?

Comparison between wireless and wired communication

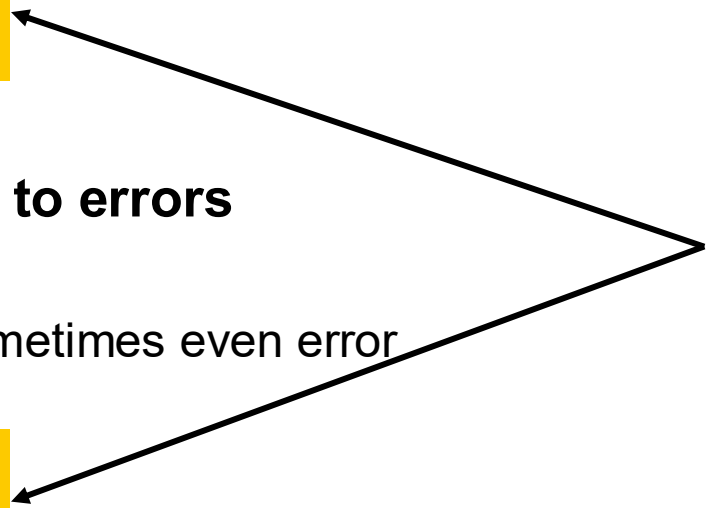
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Wired communication relies on higher layers for retransmissions

- Overhead not justified due to the low number of failed transmissions
- Noticing a corrupted packet using a checksum is sufficient
- Acceptable Bit Error Rate (BER): 10^{-12}
- On average, a failure occurs approx. every 116TB

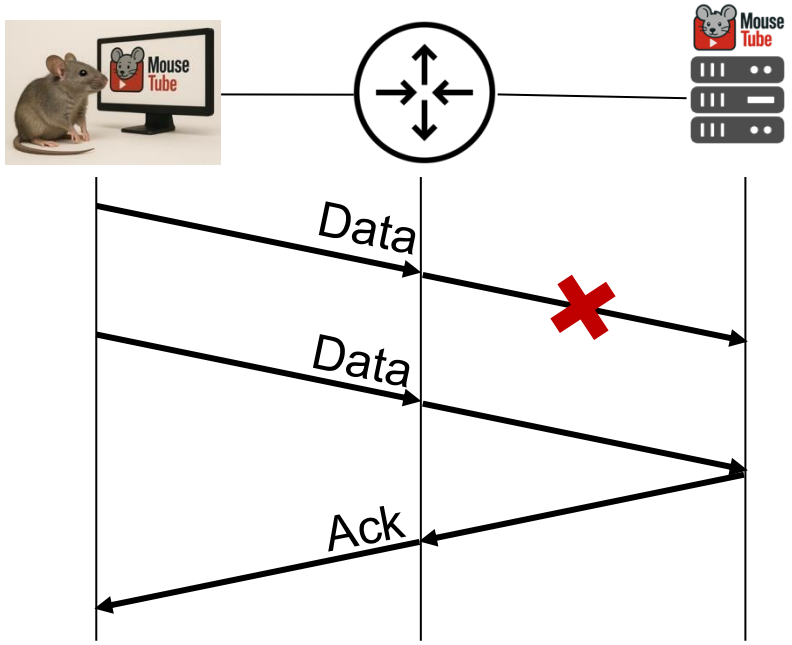
Wireless communication is much more prone to errors

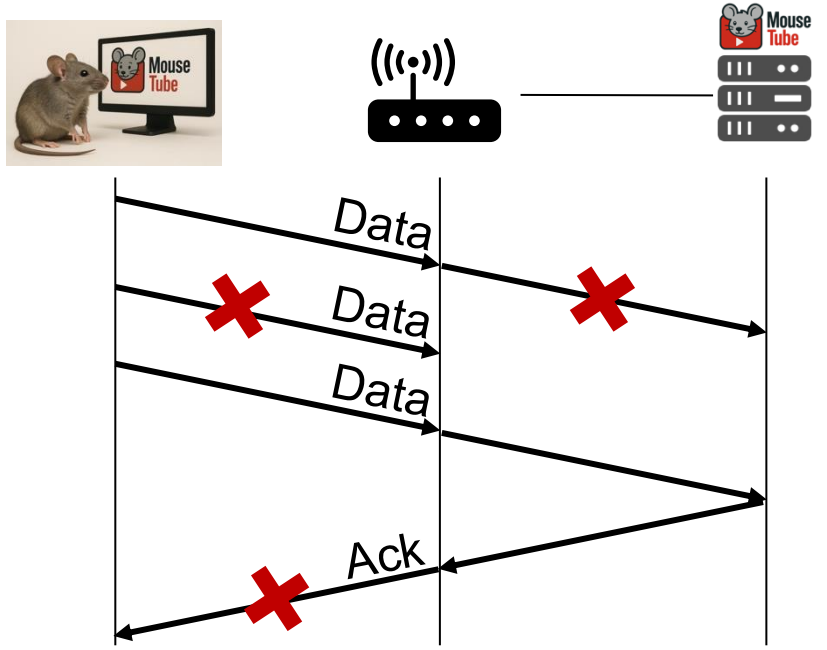
- Disturbance by other transmissions in the network
- Often requires MAC-layer retransmissions (and sometimes even error correcting codes) to function properly
- Acceptable Bit Error Rate (BER): 10^{-5}
- On average, a failure occurs approx. every 12.2KB



[1] https://www.eetimes.com/understanding-wireless-lan-performance-trade-offs
[2] https://networkengineering.stackexchange.com/questions/67473/what-is-an-acceptable-bit-error-ratio

Influence of High Bit Error Rates





Just because of the unreliable wireless link, a lot of packets are retransmitted

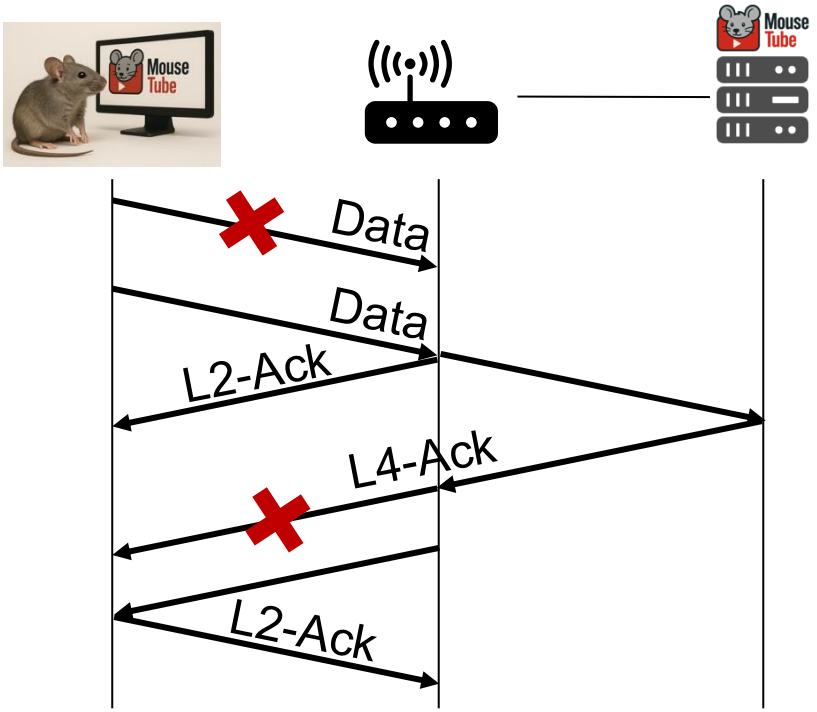
Automatic Repeat reQuest (ARQ)

Approach for reliable transmission over an unreliable channel

- Error-control method for data transmissions
- Relies on acknowledgements and timeouts to achieve ”reliable” data transmission
- TCP uses a form of ARQ to achieve reliable transmissions

Wifi uses ARQ on MAC-layer

- Relies on additional L2-acknowledgements to increase the reliability of the wireless channel
- Periodic retransmission are performed until the packet is successfully transmitted via the wireless link



It is not exactly a redundancy mechanisms,
but tries to enforce reliability on multiple layers

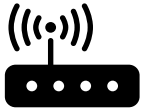
Automatic Repeat reQuest (ARQ)

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Several Versions of ARQ are available

- Stop-and-wait ARQ
- Go-Back-N ARQ
- Selective Repeat ARQ
- Hybrid ARQ (HARQ)
  - Simple HARQ
  - Chase combining
  - Incremental redundancy





Common applications

- Wifi uses stop-and-wait ARQ (pre version 7) or HARQ (version 7)
- Cellular uses HARQ (with some additions)
- TCP uses Go-Back-N ARQ (in ancient times) or Selective Repeat ARQ (today) → just FYI

Stop-and-wait ARQ

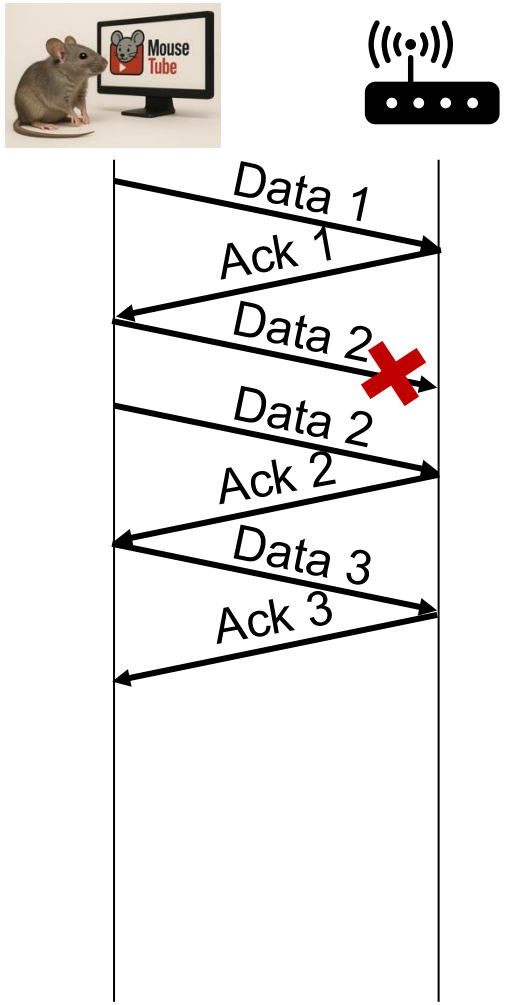
Simplest form of ARQ

- Sender transmits one frame at a time
- Only when the frame is acknowledged, the second frame is transmitted

Transmission can be very slow

- Limited by the delay of the link and the acknowledgements of the receiver
- Only usable for low-bandwidth and/or low-latency links





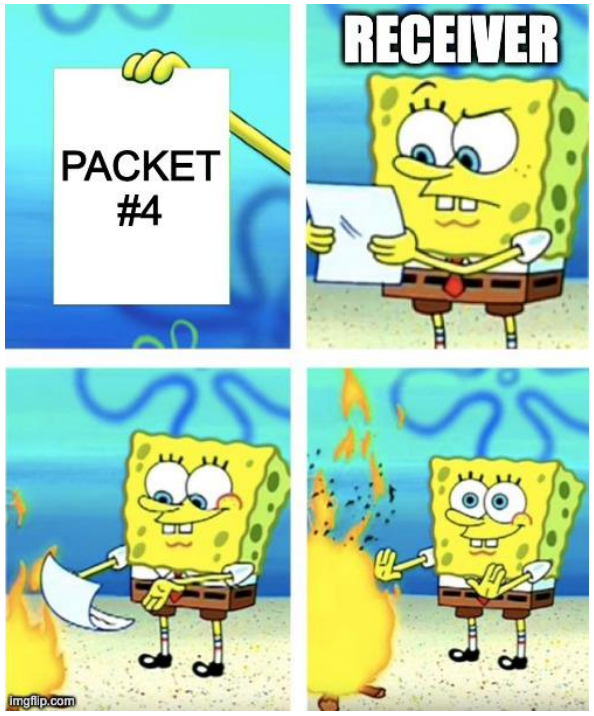
Go-Back-N ARQ

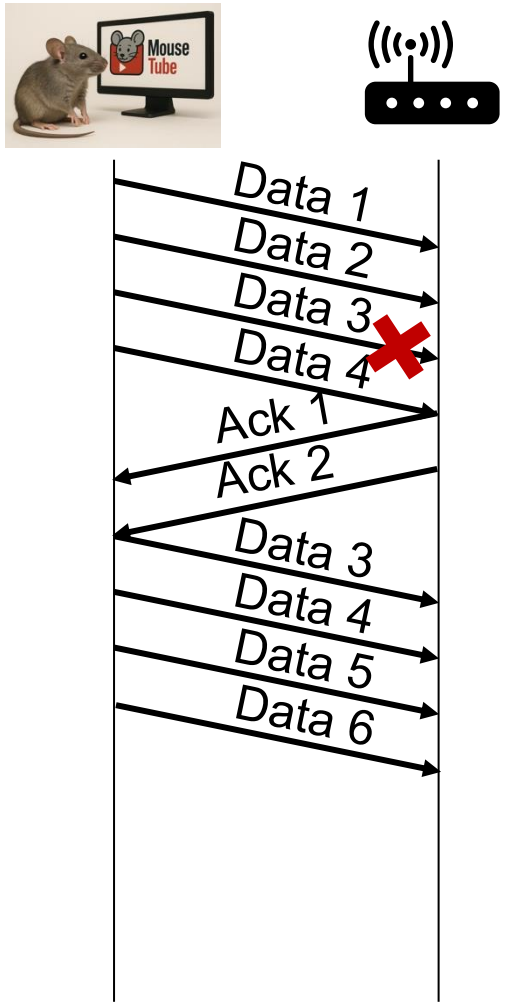
Version of ARQ used in earlier versions of TCP

- Sender transmits n-frames simultaneously
- The receiver acknowledges the last frame that was successfully received
- The sender then retransmits everything starting from the first frame that remained unacknowledged (all outstanding frames)

More efficient than Stop-and-Wait ARQ

- Sender continues transmission while waiting for Acknowledgements
- Leads to redundant transmission of already successful transmitted frames
- Was already changed for TCP to Selective Repeat ARQ in 1996



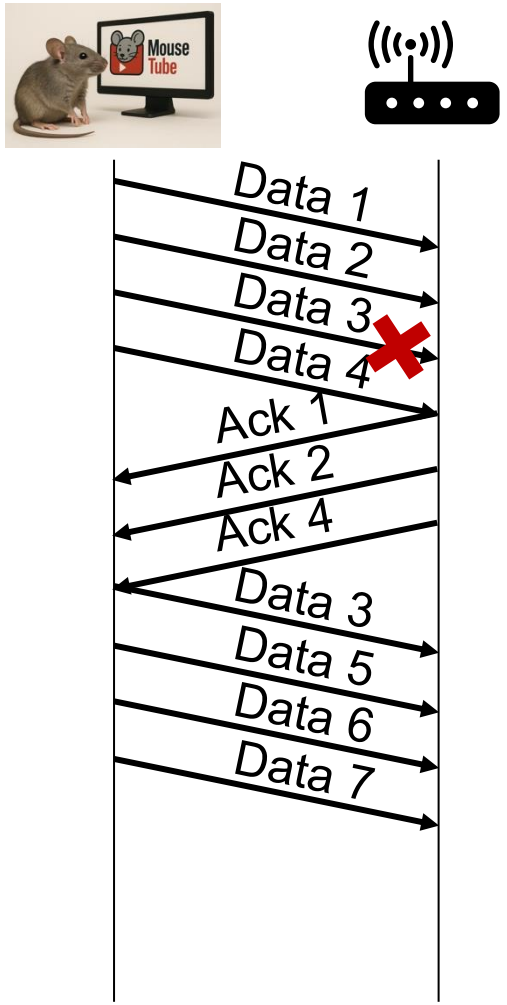


https://imgflip.com/memegenerator

Selective Repeat ARQ

Version of ARQ used in newer versions of TCP

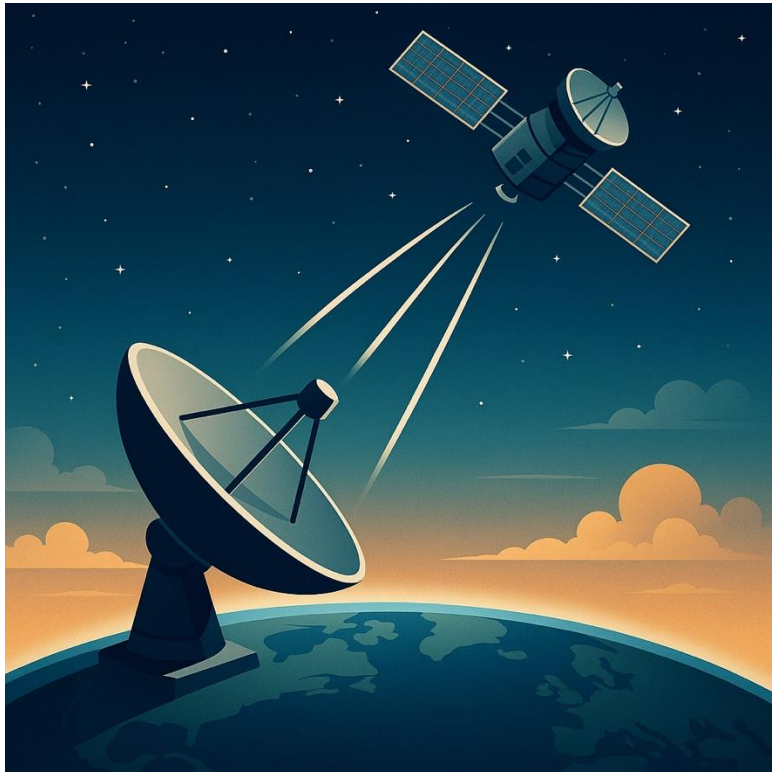
- Sender transmits n-frames simultaneously
- If at least one frame is lost, only that frame is retransmitted



Hybrid ARQ

Combination of ARQ with “something else”

- In this case, ”something else” is Forward Error Correction (FEC)
- So, instead of only using Error Detection like Cyclic Redundancy Check (CRC) for detecting failures, FEC for fixing failures is used (sometimes together with Error Detection)



Hybrid ARQ perform better than ARQ in poor signal conditions, at the cost of additional overhead

- Simple HARQ always adds FEC to each message and uses the added information to correct the message
- Chase Combining considers the message and its retransmissions in the error correction, making it more robust
- Incremental Redundancy dynamically adjusts the redundancy added, i.e., how much data volume is spent on FEC

Afaqui, M. S., Finnegan, J., & Brown, S. (2022). Evaluation of HARQ for improved link efficiency within dense IEEE 802.11 networks. Computer Communications, 191, 217-232.
G. Pan, J. Ye, Y. Tian and M. -S. Alouini, "On HARQ Schemes in Satellite-Terrestrial Transmissions," in IEEE Transactions on Wireless Communications, vol. 19, no. 12, pp. 7998-8010, Dec. 2020

Forward Error Correction with 2 Parity Bits

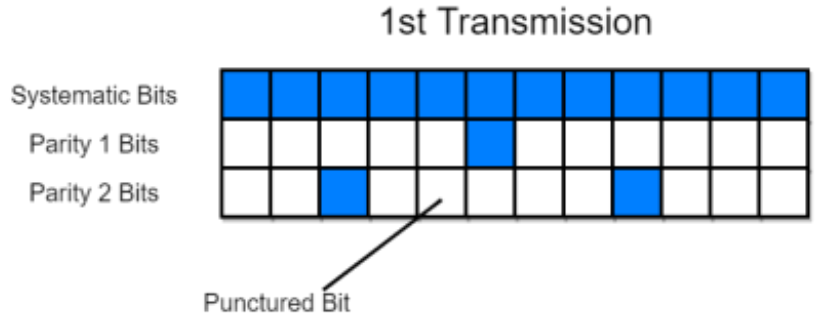
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Instead of only sending the bits of the message, additional bits are sent

- If we assume an encoder that transmits every bit using 3 bit on the physical channel (each bit is encoded using a systemic, parity 1 and parity 2 bit). Parity 1 and 2 are for Forward Error Correction (FEC)
- When transmitted, some of the bits are punctured (left out) for higher efficiency

The parity bits are used to recover the original message

- Using known functions to recover the bits (like XOR of specific bits)
- Calculate the likelihood of a certain bit being `0` or `1`

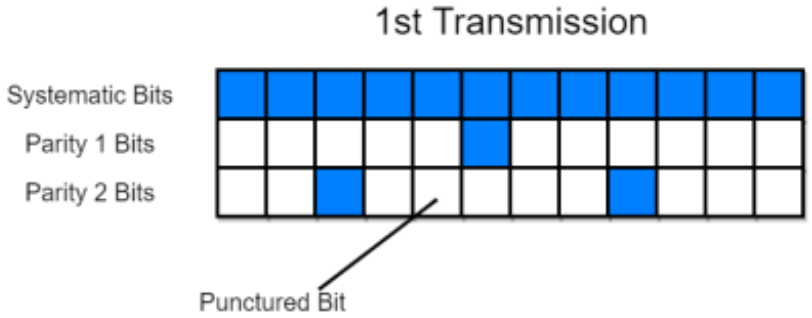


HARQ – Chase Combining

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In Chase Combining, always the same bits are sent

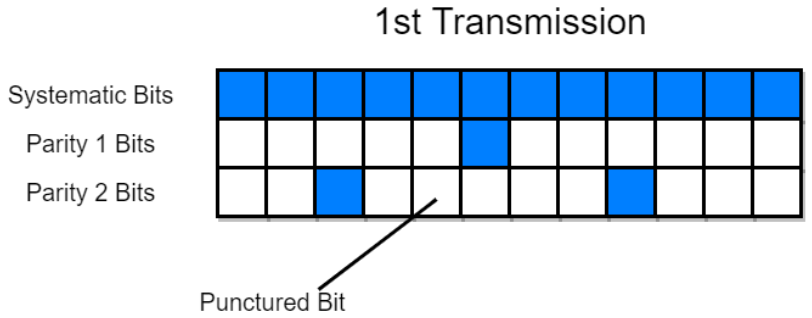
- If a message is invalid after decoding, a second message received
- By combining the two messages, the receiver can be more confident of what was actually sent



HARQ – Incremental Redundancy

In Incremental Redundancy, different bits are sent

- Each time a retransmission occurs, a different punctuation pattern is used
- By combining the two messages, the receiver has more information available for error correction, increasing the likelihood of a success



What have we learned?

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Wireless connections are (generally) less reliable than wired connections

- While there are no automated retransmissions and/or acknowledgements in Ethernet, they are frequently used in wireless connections to produce “reliable” connections
- While simple error detection is sufficient for wired connections, wireless often utilizes Forward Error Correction (FEC) to decipher messages